Date: May 29, 2026
To: Prof Nate Sniadecki, Vinh Nguyen, Zachary Lewandowski
From: Kit Nguyen
Subject: Analysis of Silicone Post for Engineered Heart Tissue (EHT)

EXECUTIVE SUMMARY

The new versions of the post used in testing the contractility of Engineered Heart Tissue is analyzed to test the effect of the base on the deflection of the post. This is done through ANSYS FEA analysis. To find the right mesh type and size would result in the most accurate model, 3 mesh types with 4 sizes for each type were tested for maximum deflection against the theoretical value obtained from the Euler-Bernoulli beam deflection equation. Then, the best element type and size was used to measure the beam deflection of the post with the base attached and the stiffness was determined. This resulted in a stiffness of 0.869 N/m for the post without the base, and 0.785 N/m for the post with the base.

INTRODUCTION

The Cell Biomechanics Lab uses silicone posts to measure the force from the contractility of engineered heart tissues (EHT). The current design is shown in **Figure 1**.

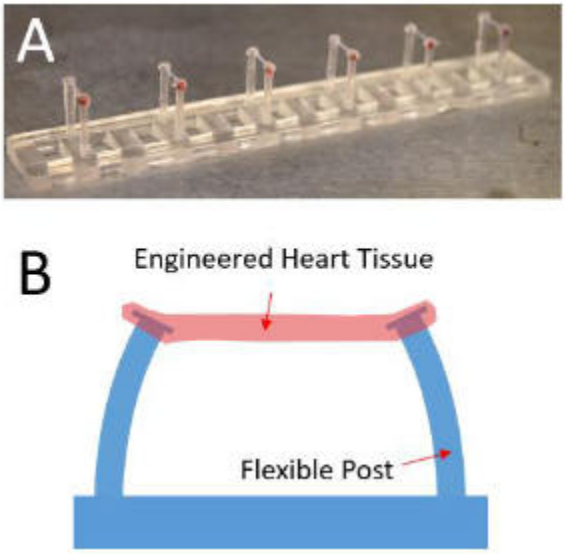


Figure 1: (A) Array of silicone posts. (B) Current design of the silicone post

A new design is considered with a hollow post **(Figure 2).** The analysis in this report is done based on the new design for one of the posts with and without the base. It is tested for the total deformation for a given force. The material that is used for post is polydimethylsiloxane (PDMS). The properties for which are shown in **Table 1**.

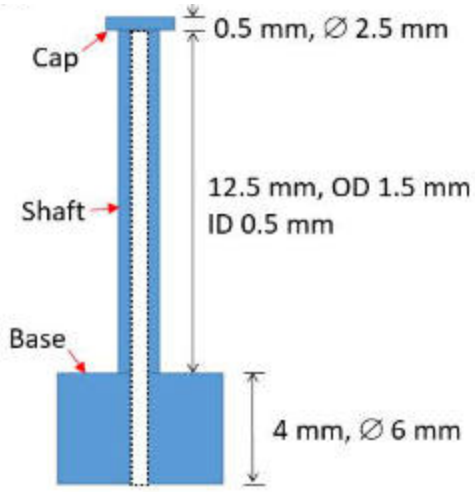


Figure 2: New proposed post design

Table 1: Properties of the PDMS

Young's Modulus (MPa)	2.5
Poisson's Ratio	0.499

METHODS

The analysis was done in 2 stages. The 1st stage was with just the post. Here, 12 different meshes were done to determine the most "accurate" mesh. These meshes were:

- Linear tetrahedral w/ element sizes of 1E-3, 5E-4, 1E-4, and 5E-5 meters
- Quadratic tetrahedral w/ element sizes 1E-3, 5E-4, 1E-4, and 5E-5 meters
- Quadratic hexahedral w/ element sizes 1E-3, 5E-4, 1E-4, and 5E-5 meters.

To determine the most "accurate" mesh, these were compared with an ideal value using Euler's Formula:

$$F = \frac{3EI}{L^3}\delta$$

With F = force, E = Young's Modulus, I = second moment of area, L = Length, and δ = deflection.

When finding the theoretical deformation from a 0.2 mN horizontal force, the assumption that the cap acts as part of the hollow cylinder (post/shaft) is used. Additionally, for this to work, the beam needs to be slender and uniform, which is true in this case. This results in $\delta = 2.387 * 10^{-4}$m, which will be used to compare to the experimental data.

Once the model was ready to be set up, boundary conditions were applied. The boundary conditions that I used for the FEA without the base was a fixed boundary condition at the bottom of the post and the horizontal force applied to the entire cap as shown below in **Figure 3.** This represents the system where the entire cap is covered by the EHT, therefore the force acts on the entire cap as it is pulled. The base is a fixed support because that would represent the base, where it doesn't move from.

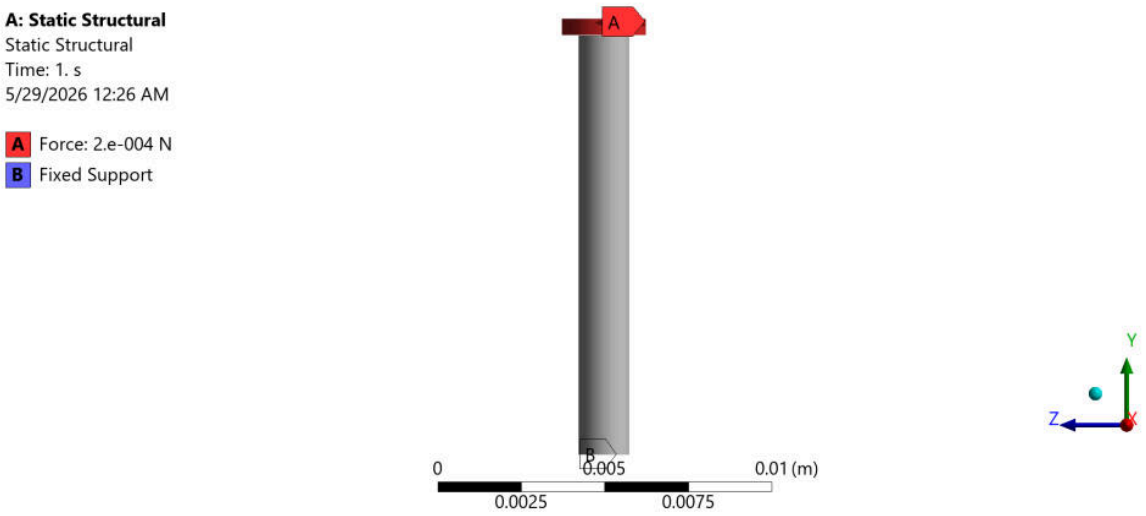


Figure 3: Post Boundary Conditions

Another version of this configuration was looked at where the EHT would only pull on part of the endcap, representing a more accurate model where the EHT grabs the back end of the endcap and pulls it **(Figure 4)**. This model is more accurate because it represents where the force typically would act on the post since it is not really pulling on the front face of it. This, however, resulted in percent errors of >> 1%, so it was not used. The meshes, deformations, and table for this model is shown in **Appendix 5.**

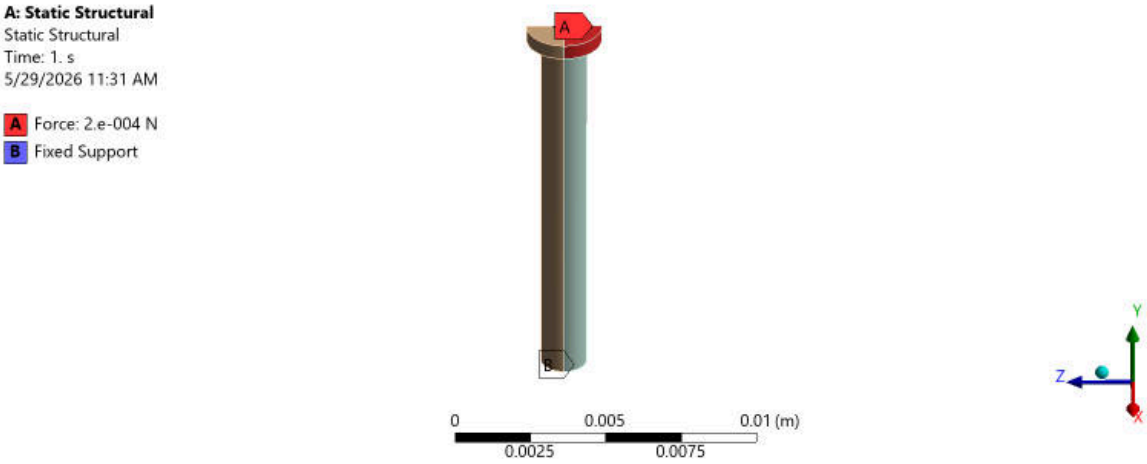


Figure 4: Post Boundary Conditions on a more accurate model

Once the mesh type and size was found, stage 2 of the analysis can be done. Stage 2 of the analysis is with the base and using the best element type and size.

RESULTS

Stage 1

The summary of stage 1 for the different meshes in this analysis is shown in **Table 2** (meshes and FEA results are shown in **Appendix 1 & 2**). The maximum horizontal deflection was from a 0.2 mN Horizontal force.

Table 2: Percent error from different meshes at 0.2 mN

Element Type	Element Size (m)	Number of Elements	Maximum Horizontal Deflection (m)	Percent Error	Notes
Tet4	0.001	617	0.000052296	78.09165431	
Tet4	0.0005	1523	0.000059912	74.90108599	
Tet4	0.0001	47905	0.00015121	36.65364556	
Tet4	0.00005	193429	0.00017086	28.42167767	
Tet10	0.001	643	0.00024304	1.816665445	
Tet10	0.0005	1542	0.00023124	3.126704586	
Tet10	0.0001	46696	0.00023242	2.632367582	
Tet10	0.00005	189381	0.0002329	2.431281344	
Hex20	0.001	1241	0.00022988	3.696448928	
Hex20	0.0005	2183	0.00023077	3.323601527	
Hex20	0.0001	24714	0.00023276	2.489931497	
Hex20	0.00005			100	FAIL
Hex20	0.000052	212650	0.00023305	2.368441894	

	0.000052		0.00025474	6.718142509	

Shown above in **Table 1** is the data from the different meshes. For each mesh, the element types, element sizes, number of elements, maximum horizontal deflection, and percent error were taken, with the percent error coming from the theoretical value: $Percent\ Error\ (\%) = (|Experimental - Theoretical| \div Theoretical) \times 100$. For the 0.00005m Hex20 mesh, the mesh failed so it could not be tested.

The lowest percentage error with the least elements was from the Tet10 0.001m mesh. Since this was the lowest percentage error to the theoretical value, this mesh type and size was used for the rest of the analysis. The percentage error for this mesh was 1.82% with 643 elements. Although it does not reach 1% of the theoretical value, it also needs to be noted that the theoretical value made the assumption that the cap acted as part of the post, so it might not be as accurate.

Using the most accurate mesh type as a base, more models were made to determine the stiffness of the PDMS. To calculate the spring constant of the post (without the base), the total deformation is plotted against different forces acting on the post ($F = k\delta$) for forces of 0, 0.1, 0.2, 0.3, and 0.4 mN. The plot is shown below **(Figure 5)**. Using the slope of the line of best fit, the spring constant is 0.869 N/m.

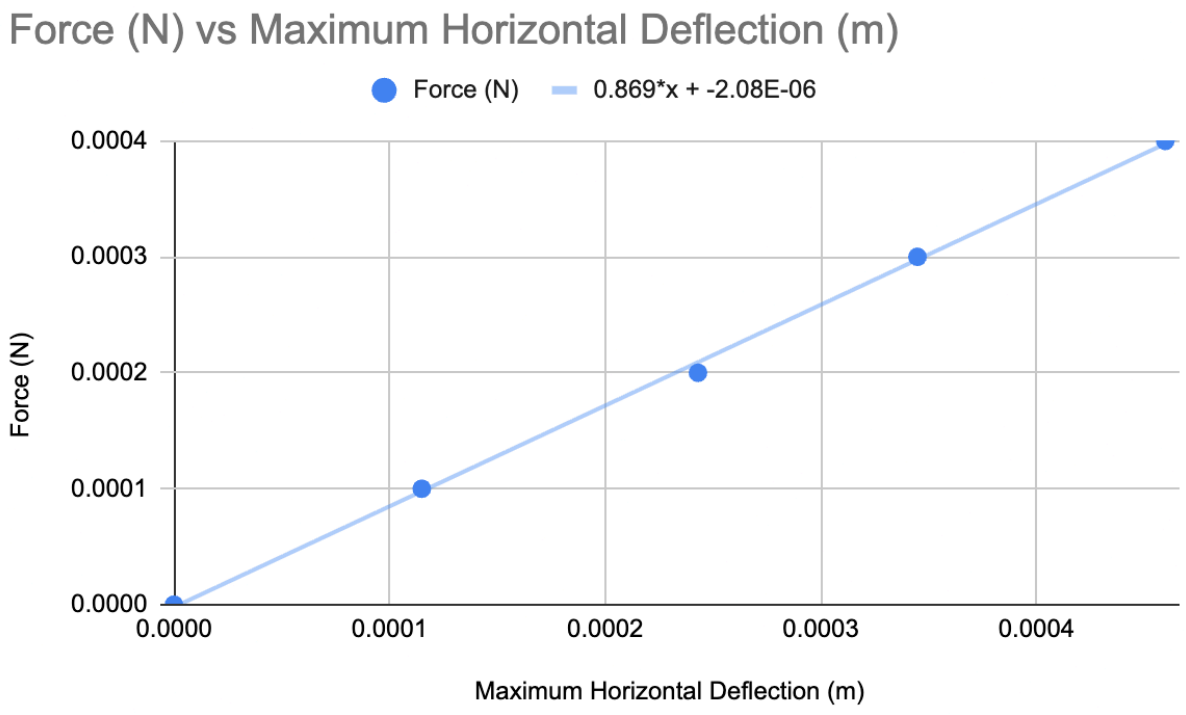


Figure 5: Force vs Maximum Horizontal Deflection Plot for Stiffness Calculation **(**Mesh and models in **Appendix 3)**

Stage 2

Similarly, the stiffness of the base with the post was calculated using the same method using 0, 0.1, 0.2, 0.3, and 0.4 mN. Setup for the base is shown in **Figure 6**. The boundary conditions for this are similar to the post without the base with the force being applied to the whole cap. This time, the fixed support goes at the bottom of the base. This is done to represent the base also being on the tray. Since the base is the same material and also affected by the EHT, its deflection needs to be accounted for as well. The results are shown in **Figure 7**.

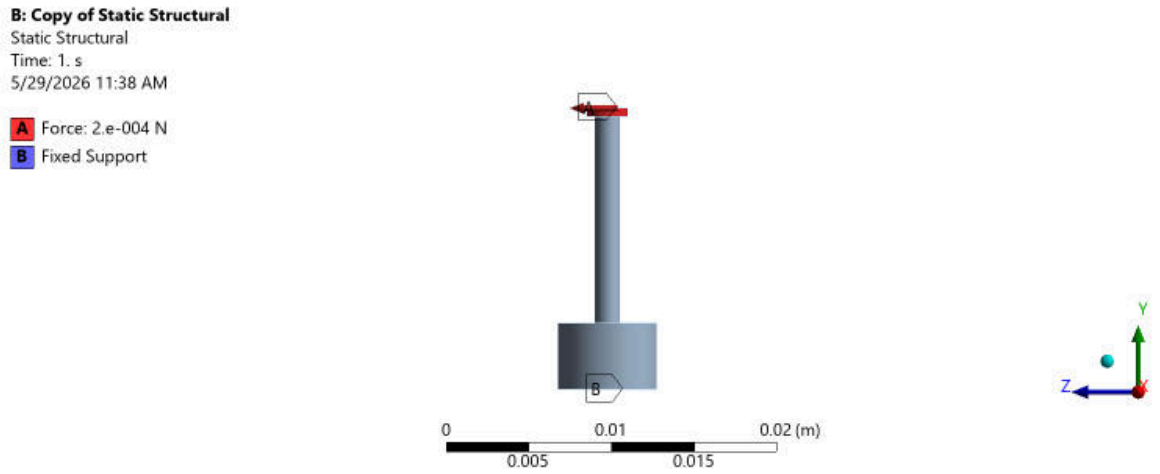


Figure 6: Post Boundary Conditions with Base

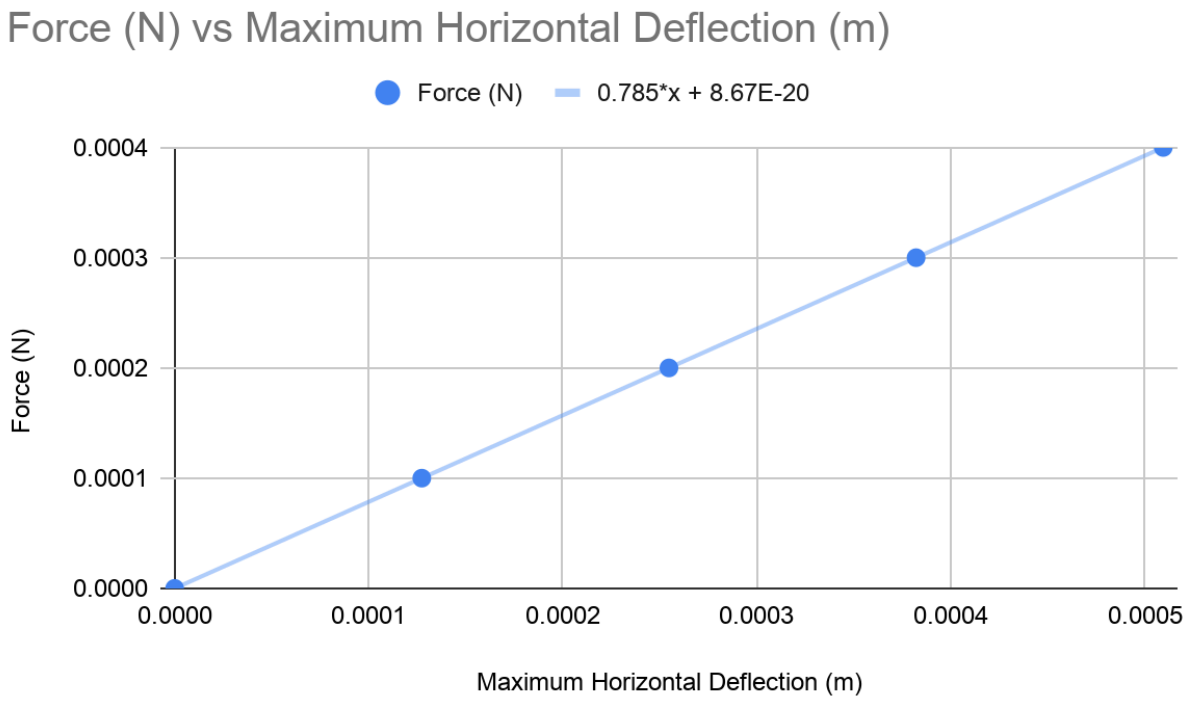


Figure 7: Force vs Maximum Horizontal Deflection with Base Plot for Stiffness Calculation **(**Mesh and models in **Appendix 4)**

The plot shows that there is a slight decrease in the stiffness of the post once the base is added from 0.869 N/m to 0.785 N/m. This decrease in stiffness was the result of more deflection happening at the tip of the endcap.

CONCLUSION

The results from these analysis' show that although the Euler-Bernoulli beam deflection is a good theoretical representation of the posts used in the Cell Biomechanics Lab, it is a simplified version that does not accurately take into account other variables and geometry.

Decreased stiffness makes sense when the base is added. When compared to the theoretical deflection, this results in a percentage error of 6.72% compared to the 1.82% from the one without the base. This is because now, the fixed support goes to the bottom of the base, so the deflection in the actual base (shown in **Figure 8**) is also considered in the total deformation as it amplifies through the whole structure. The connection between the base and the post has a defection of 6.4177e-7 so it does not account for the full extra effects of deflection, it is applied through the rest of the shaft, which is not accounted for in the Euler-Bernoulli beam deflection equation.

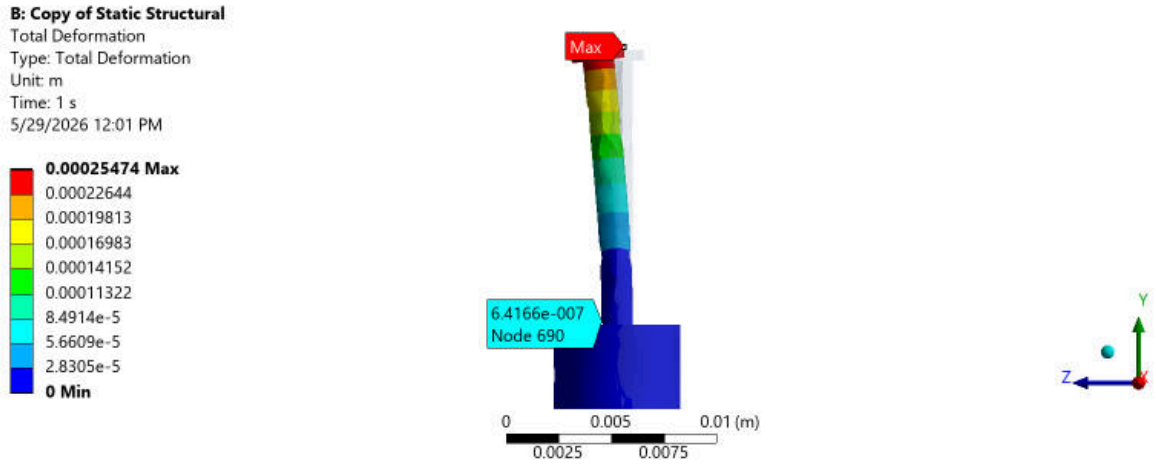


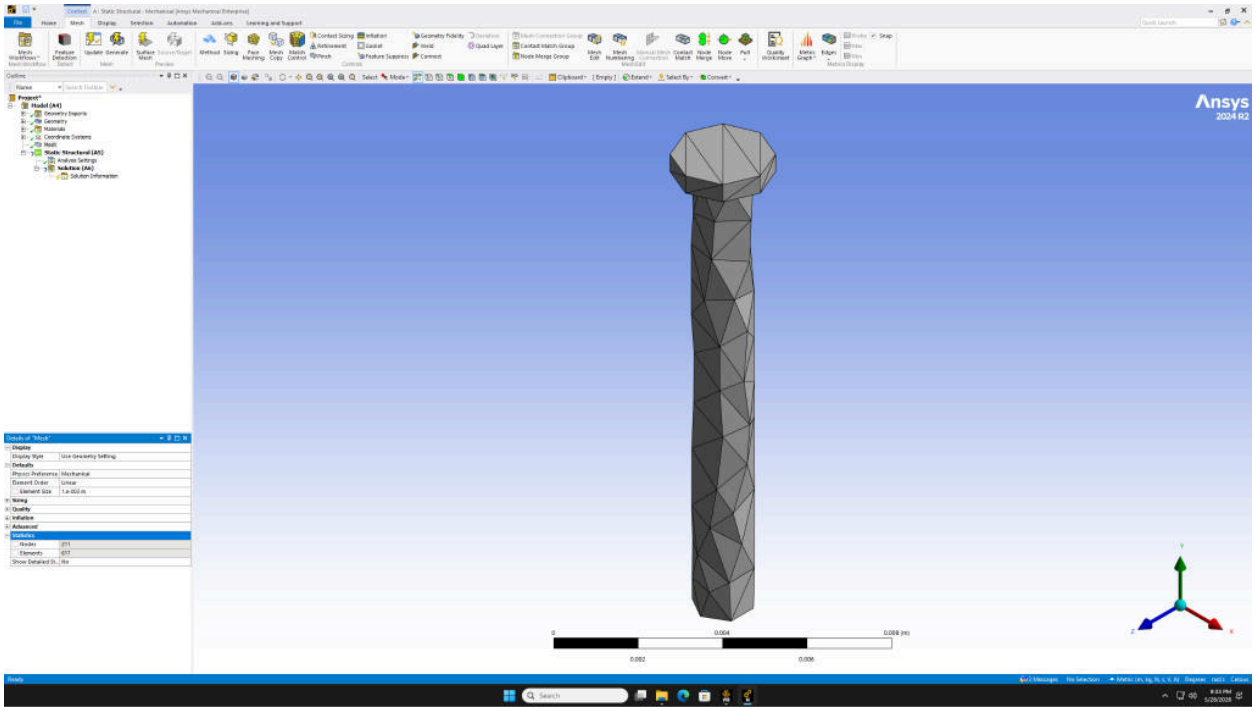
Figure 8: Total deformation of post with base

Another interesting thing about this analysis was that the "best" element type and size was the Tet10 (quadratic tets) at 0.001m element size. Normally, it would make sense to have the best/most accurate results be when the element size is the smallest, but this was not the case. A possible reason for this is because the ANSYS model deflections were compared to the theoretical deflection from the Euler-Bernoulli equation. Since the theoretical model used very loose assumptions on the geometry to simplify it, the value was higher than the real value. For

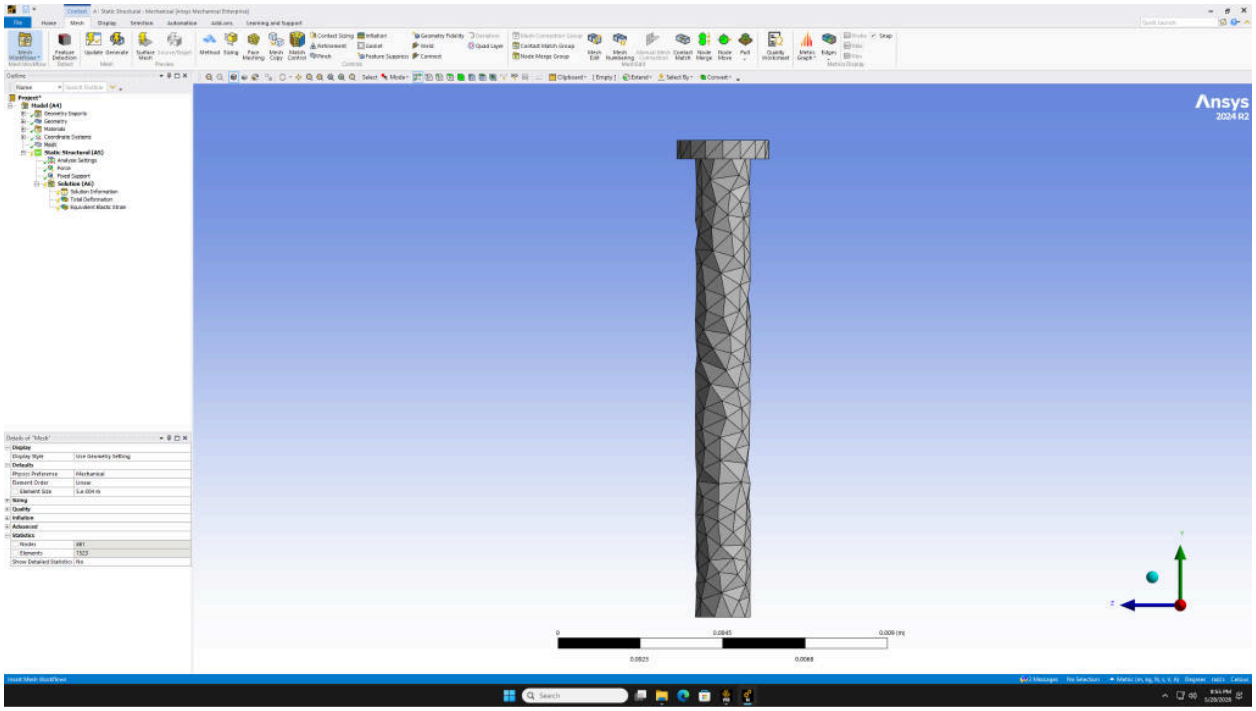
example, the assumption was made that the second moment of area was just a hollow cylinder, and the cap was not accounted for.

For the future, the theoretical value could be more accurately modeled by using more strict assumptions for geometry. Also, the model that was not used can be looked at closer. Although it did not work in this particular case, it is a good representation of how the EHT actually would pull on the post and so analysis on this can give a more accurate model that represents the physical tests. Another future recommendation is to compare these values to values of the real experiment and with the model of the original design to see which one gives more accurate data.

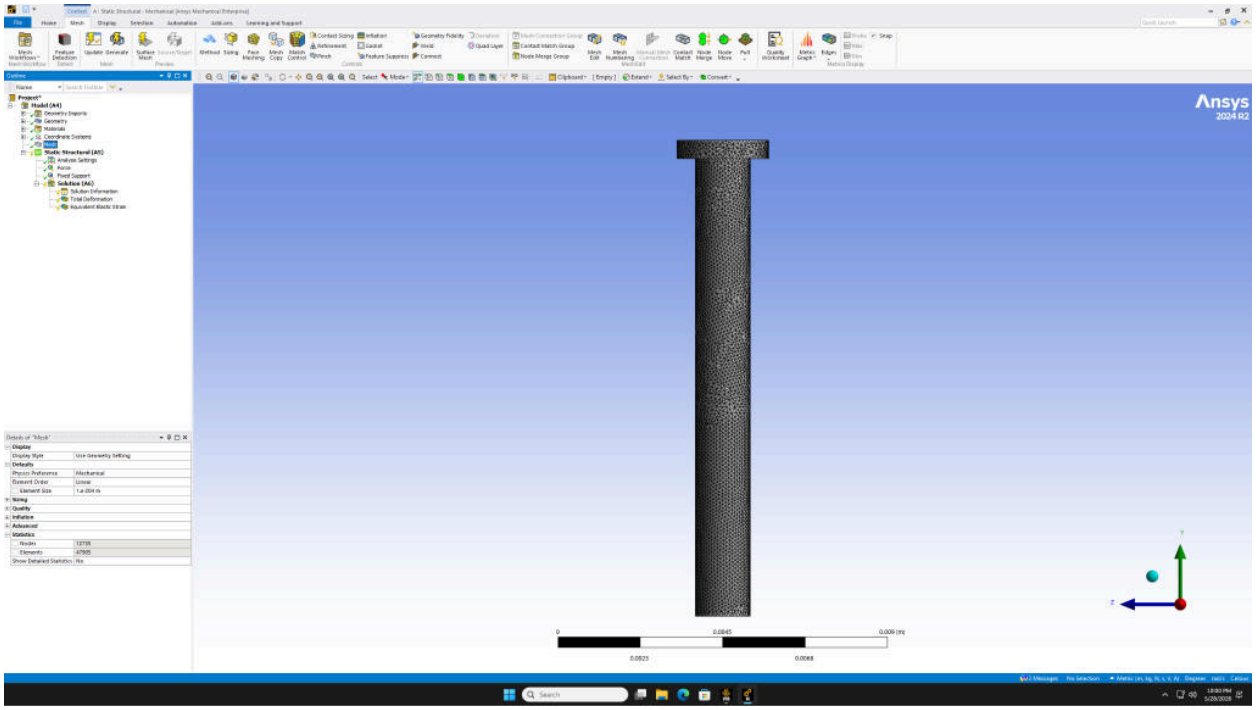
Appendix 1: Meshing



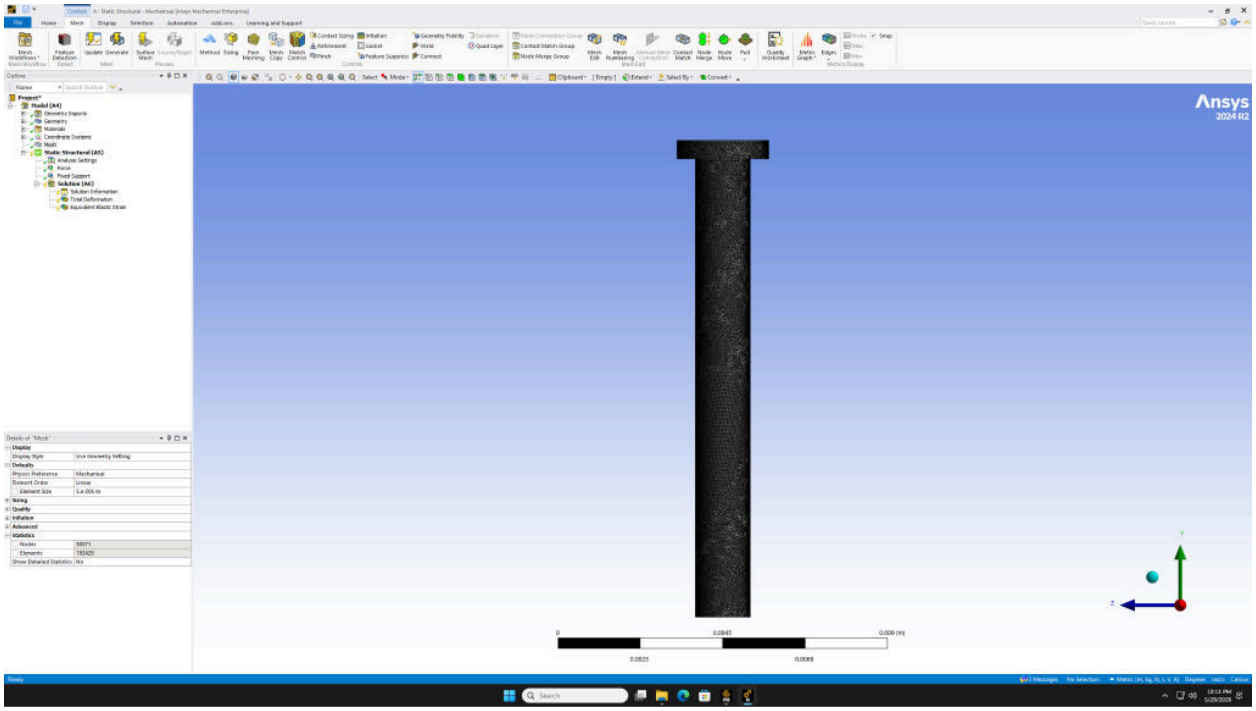
Appendix Figure 1a: 0.001m Tet4 Mesh



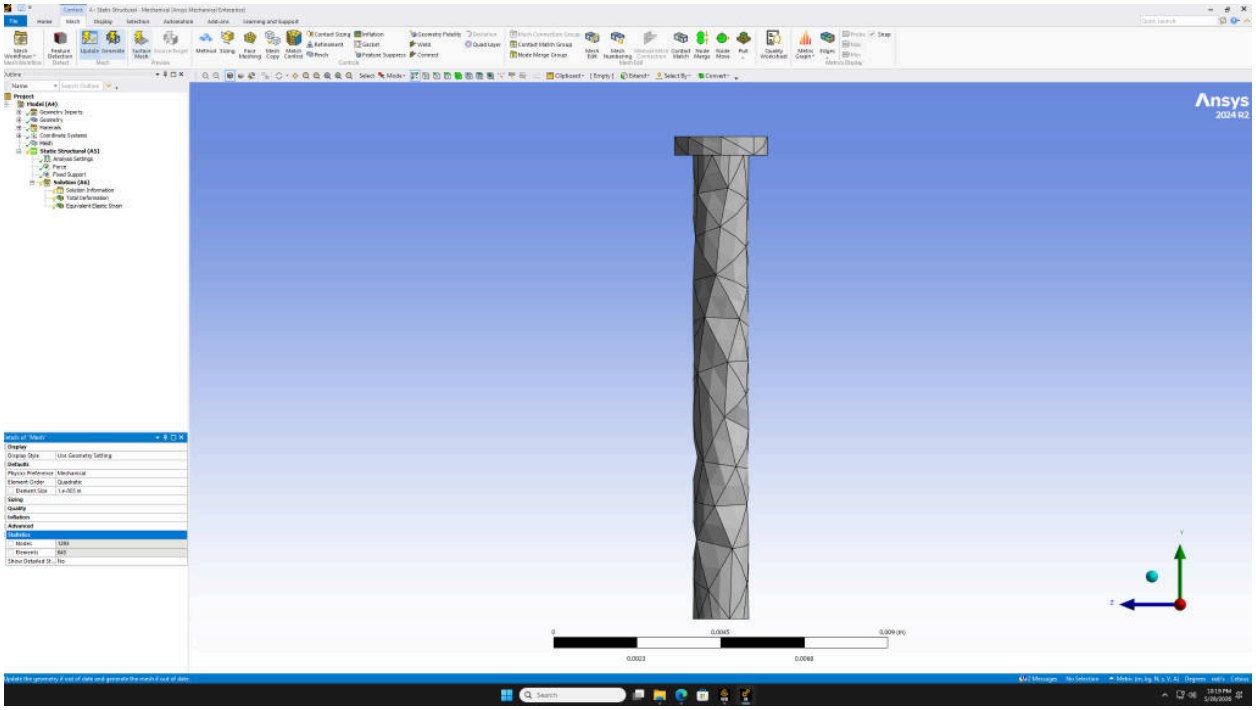
Appendix Figure 1b: 0.0005m Tet4 Mesh



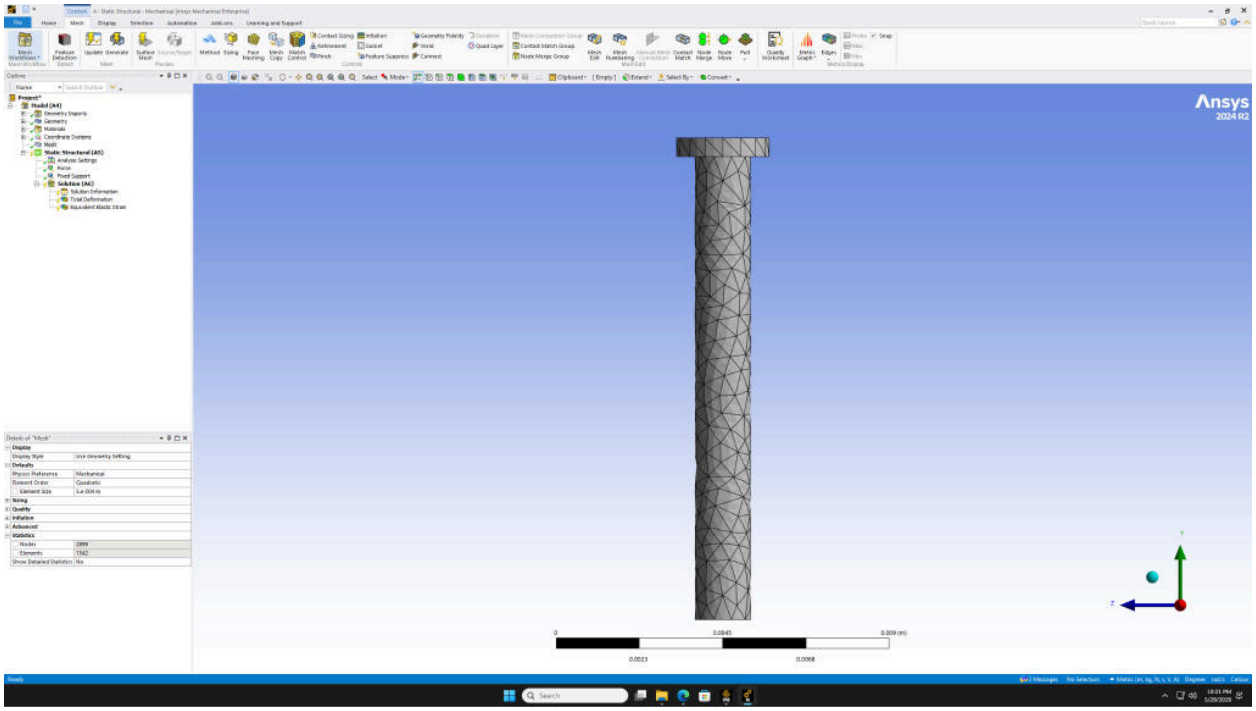
Appendix Figure 1c: 0.0001m Tet4 Mesh



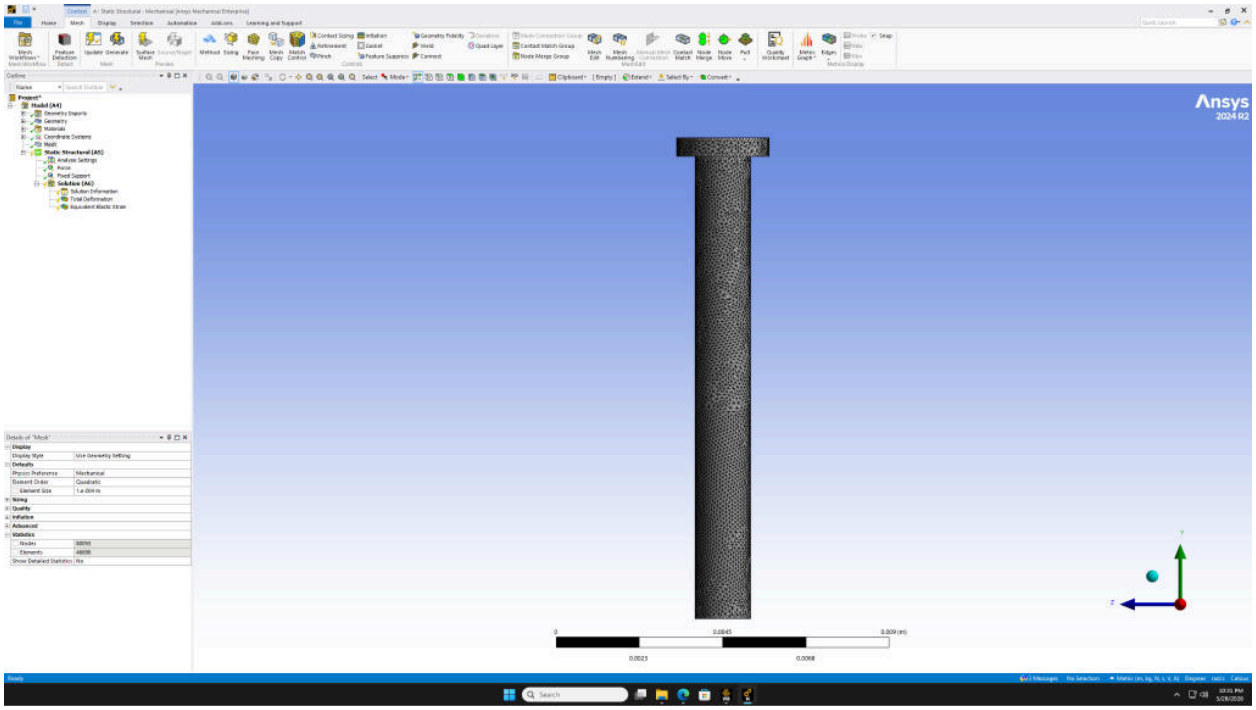
Appendix Figure 1d: 0.00005m Tet4 Mesh



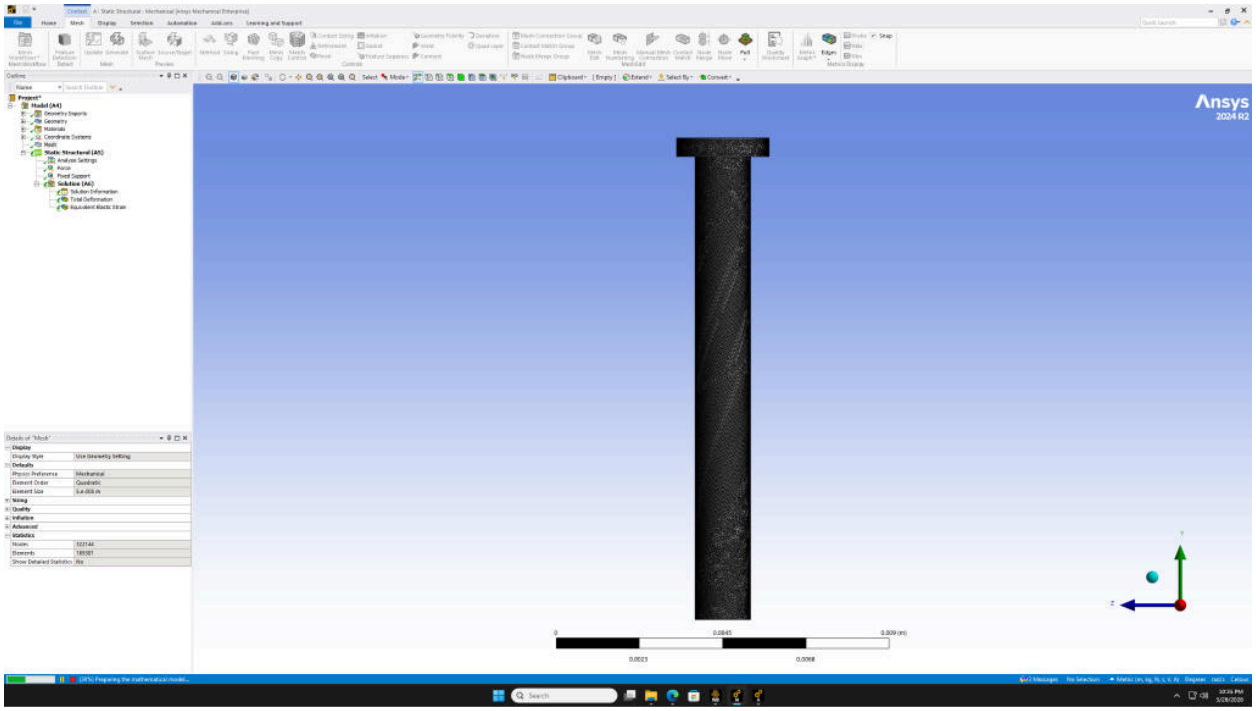
Appendix Figure 1e: 0.001m Tet10 Mesh



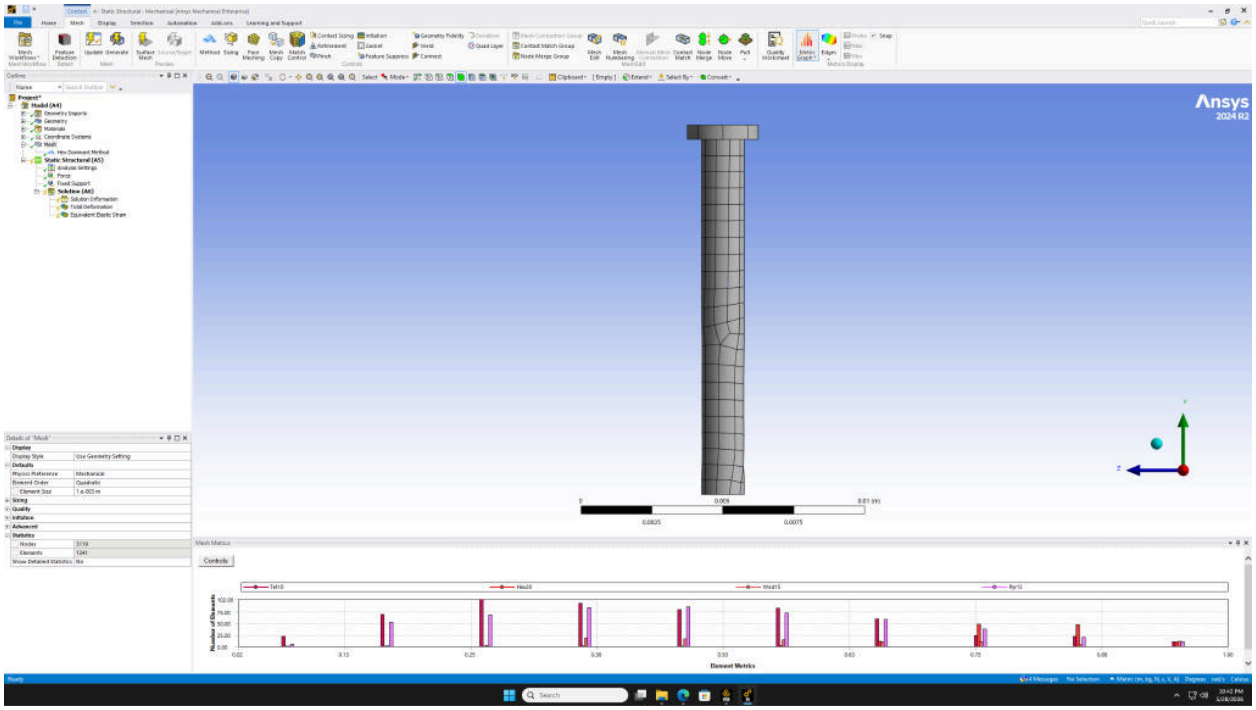
Appendix Figure 1f: 0.0005m Tet10 Mesh



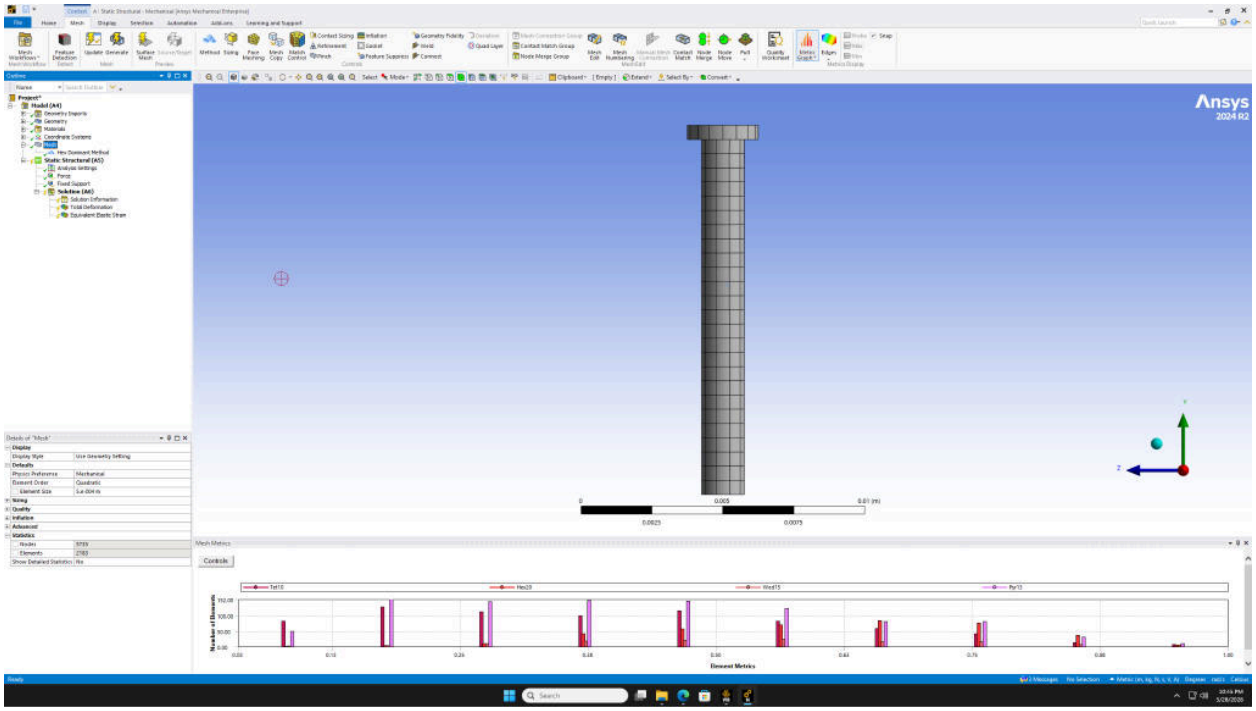
Appendix Figure 1g: 0.0001m Tet10 Mesh



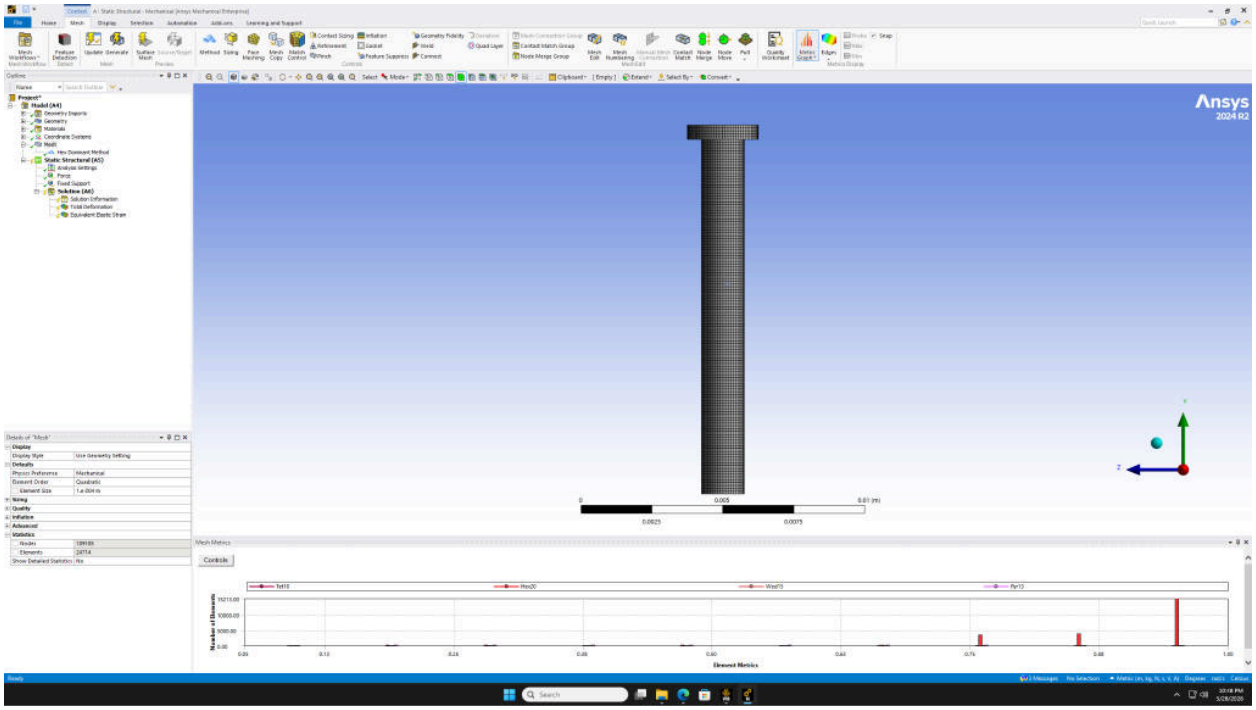
Appendix Figure 1h: 0.00005m Tet10 Mesh



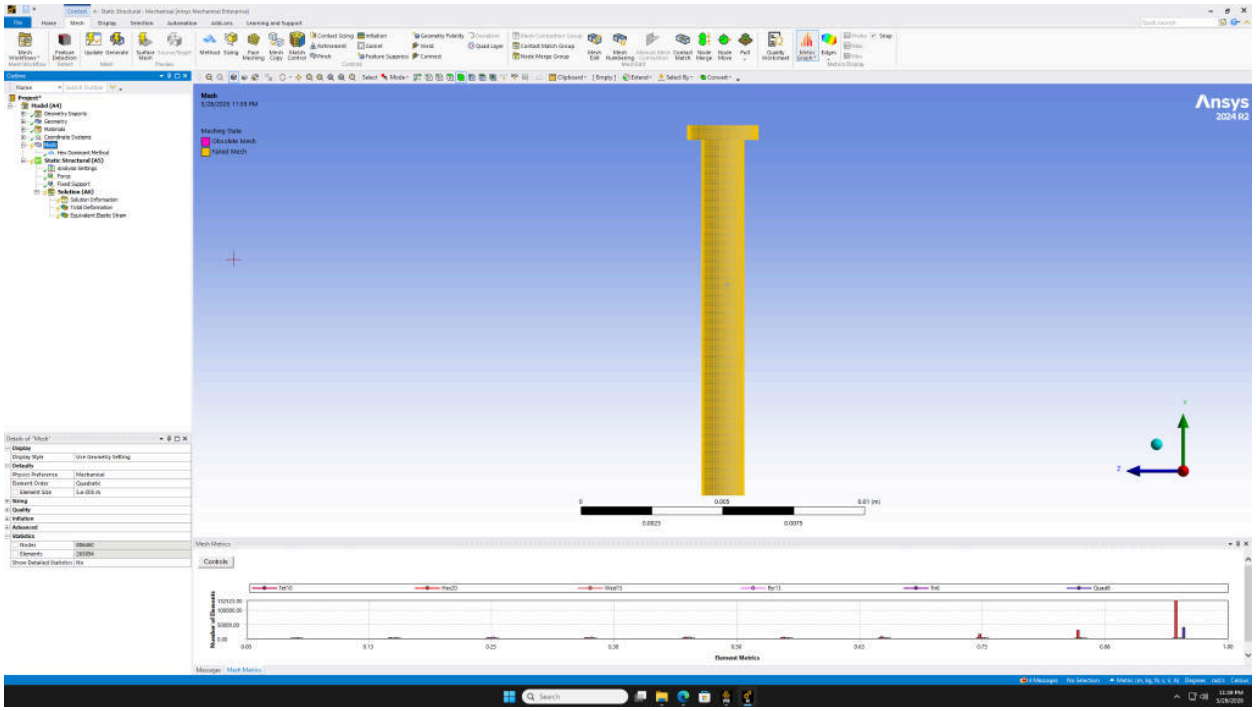
Appendix Figure 1i: 0.001m Hex20 Mesh



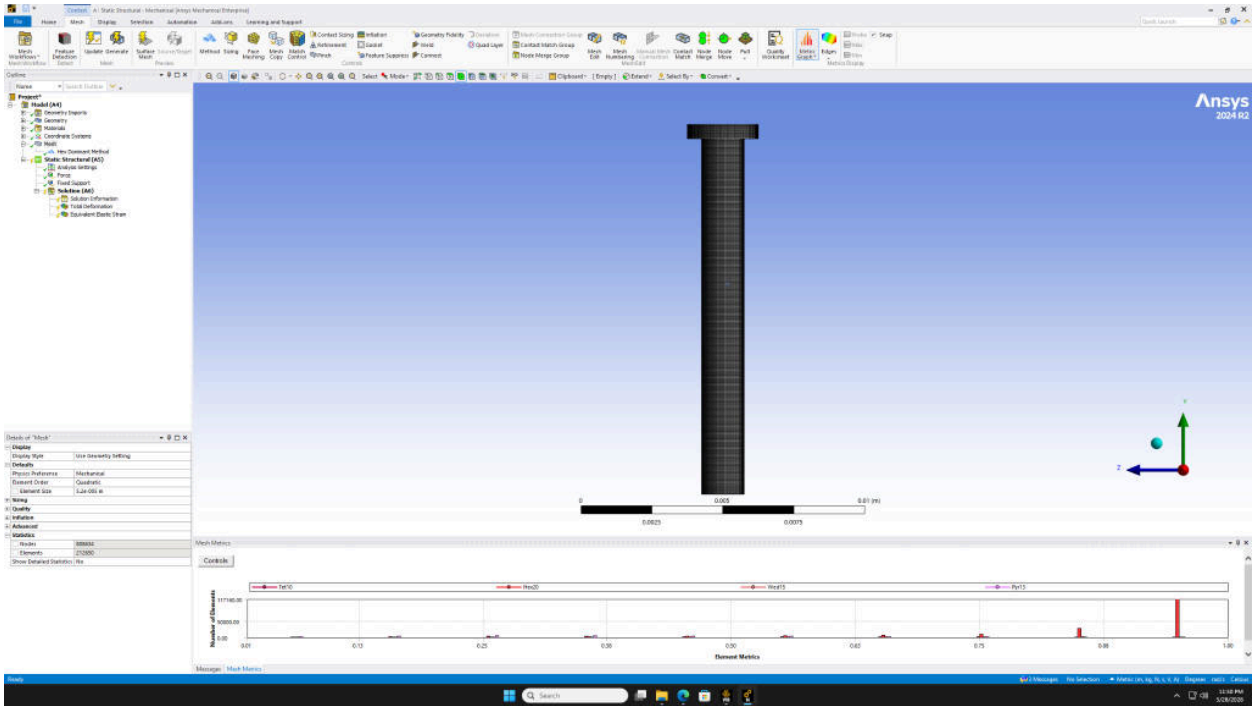
Appendix Figure 1j: 0.0005m Hex20 Mesh



Appendix Figure 1j: 0.0001m Hex20 Mesh

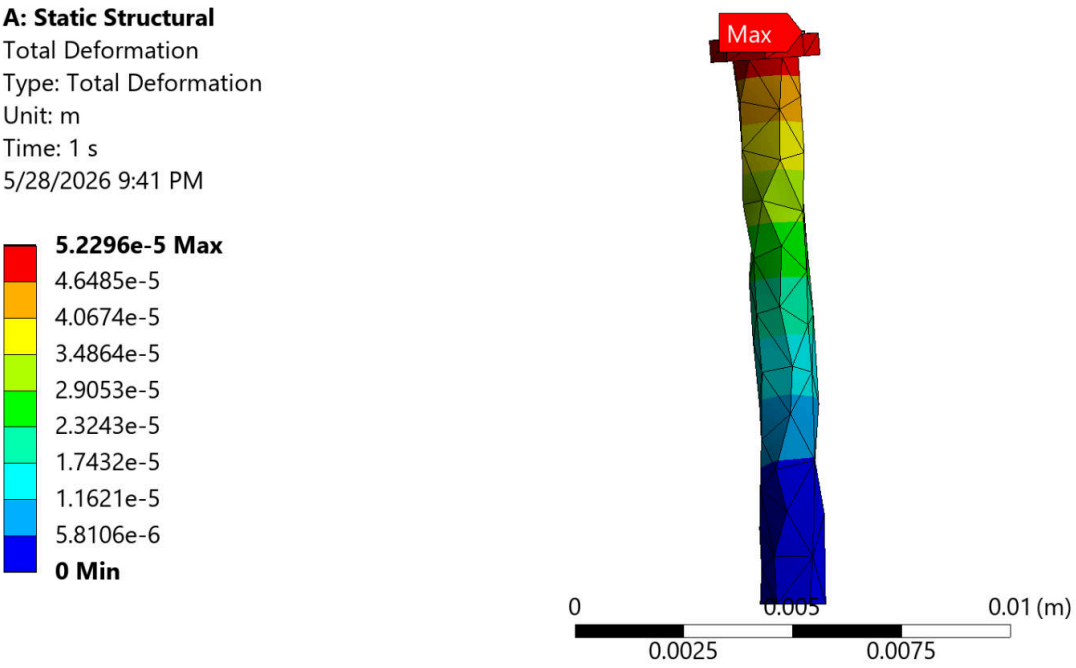


Appendix Figure 1k: 0.00005m Hex20 Failed Mesh

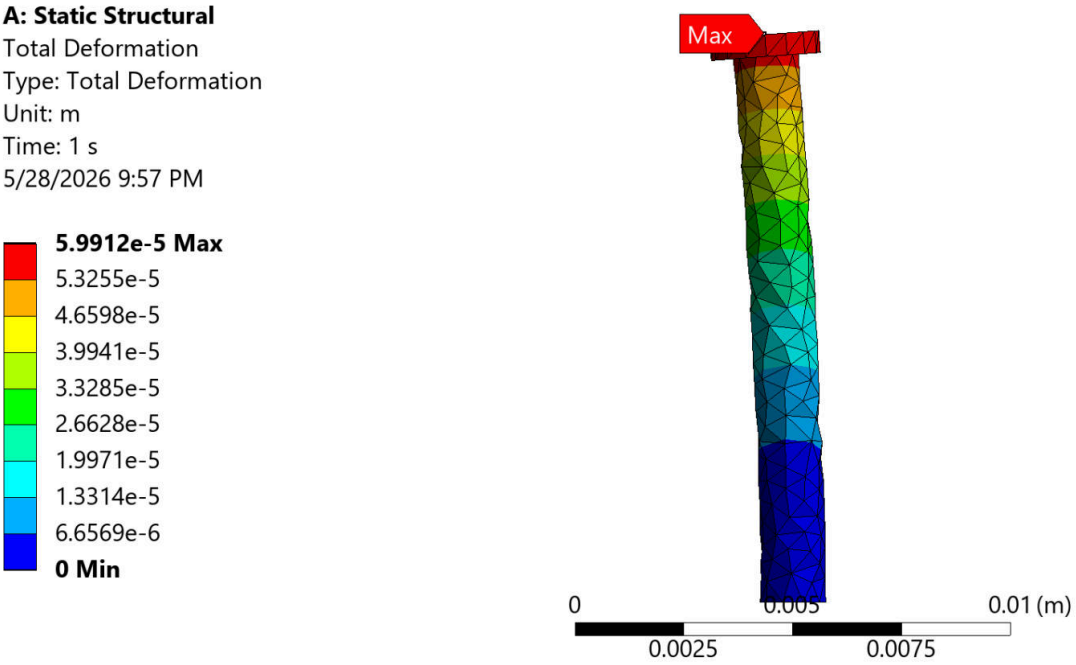


Appendix Figure 1k: 0.000052m Hex20 Mesh

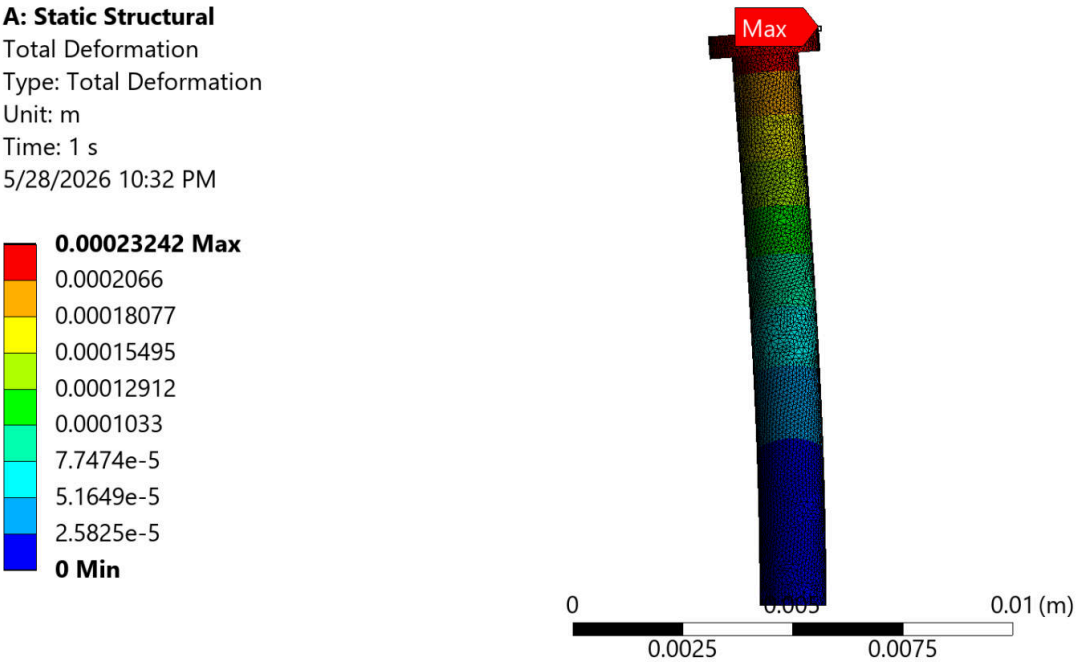
Appendix 2: Total Deformation First Stage



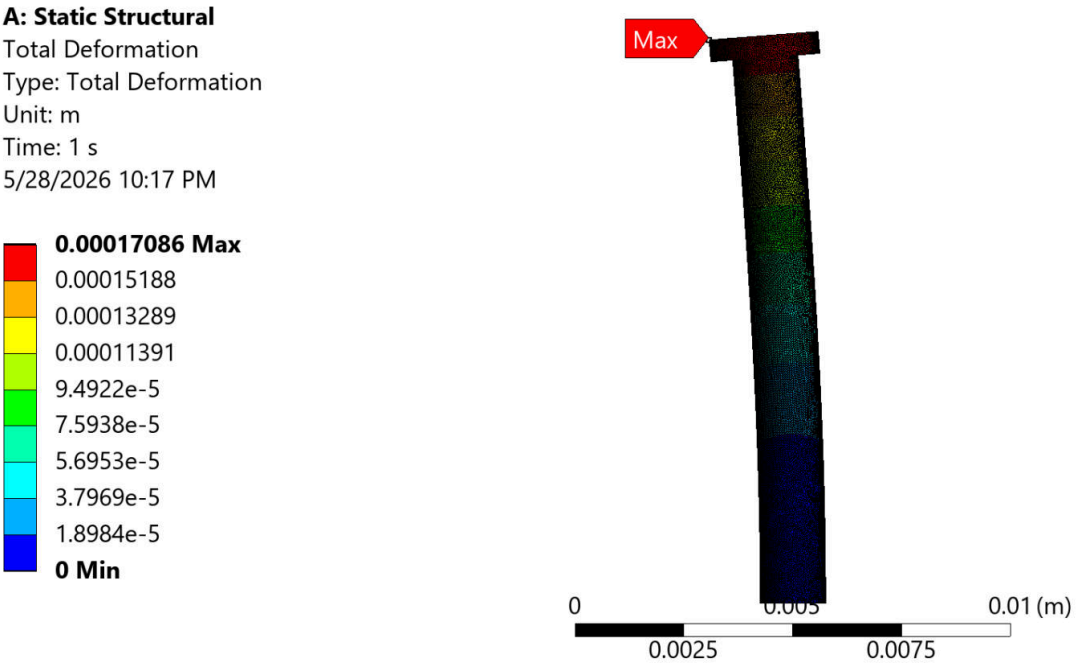
Appendix Figure 1a: 0.001m Tet4 Total Deformation



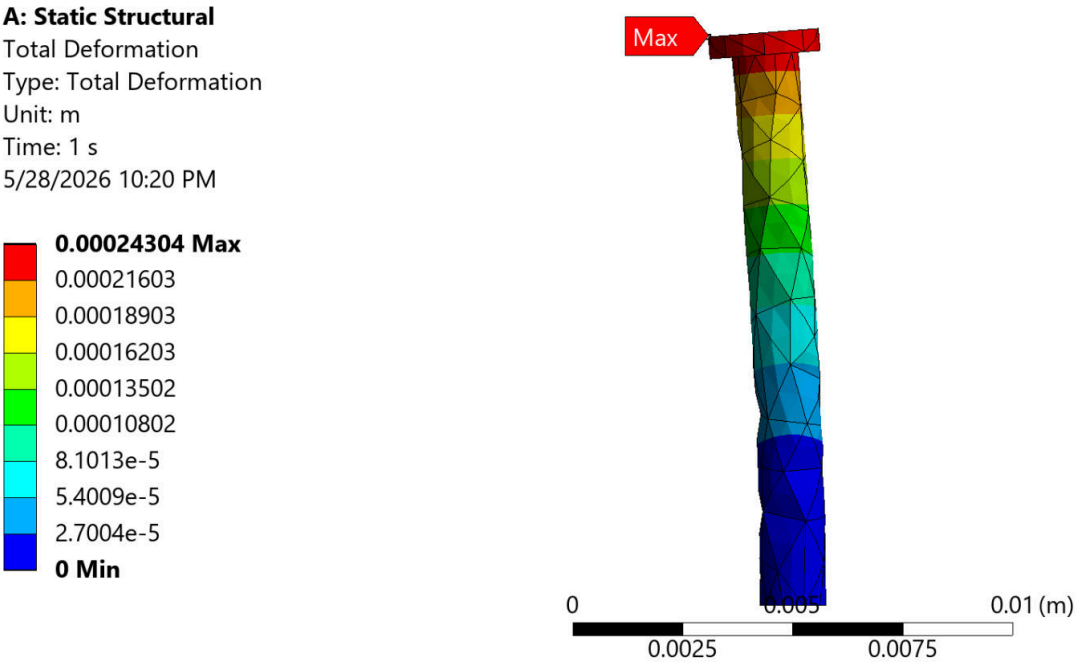
Appendix Figure 1b: 0.0005m Tet4 Total Deformation



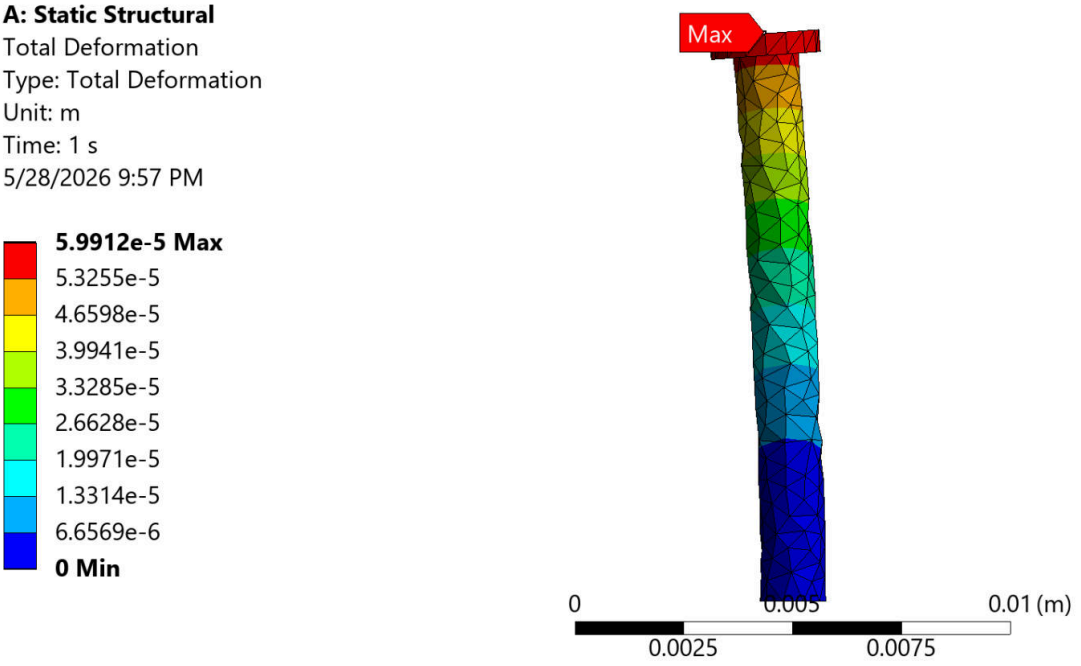
Appendix Figure 1c: 0.0001m Tet4 Total Deformation



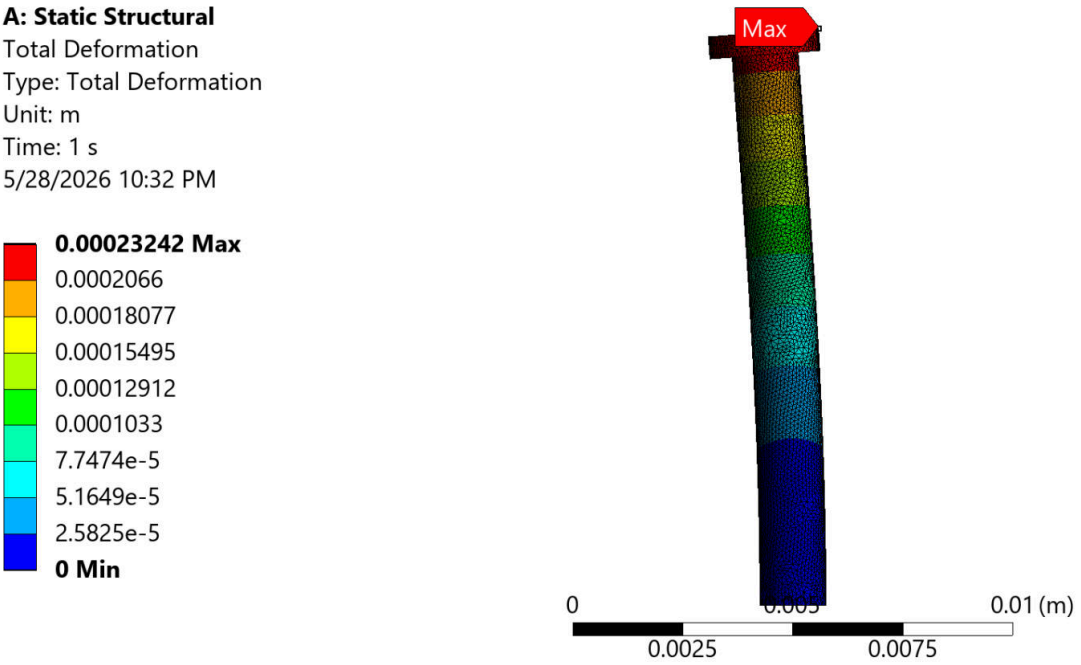
Appendix Figure 1d: 0.00005m Tet4 Total Deformation



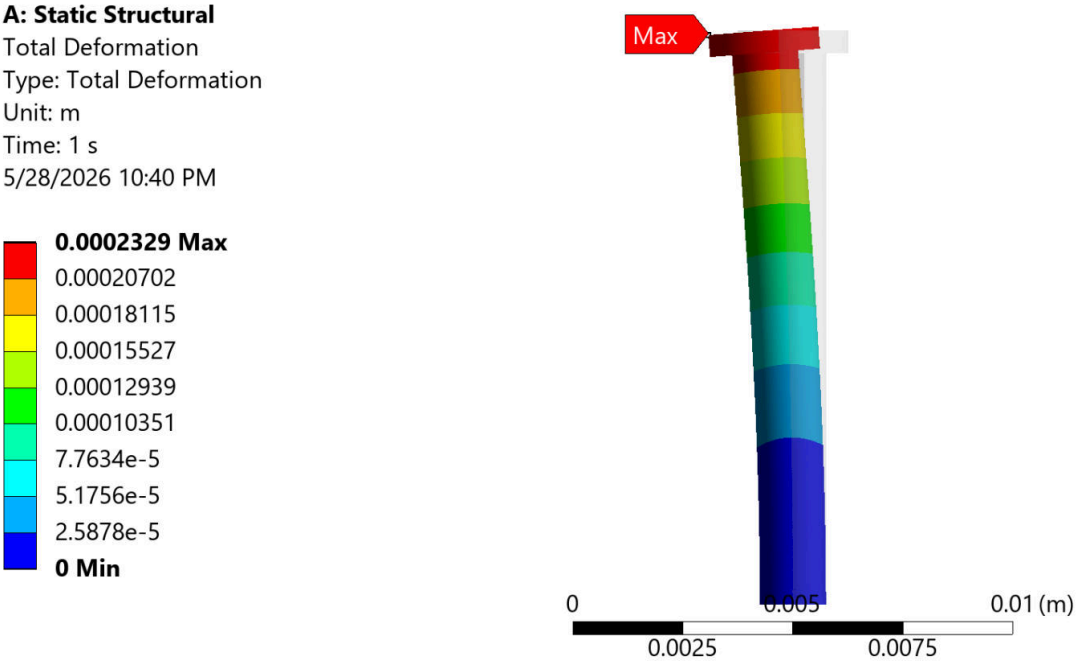
Appendix Figure 1e: 0.001m Tet10 Total Deformation



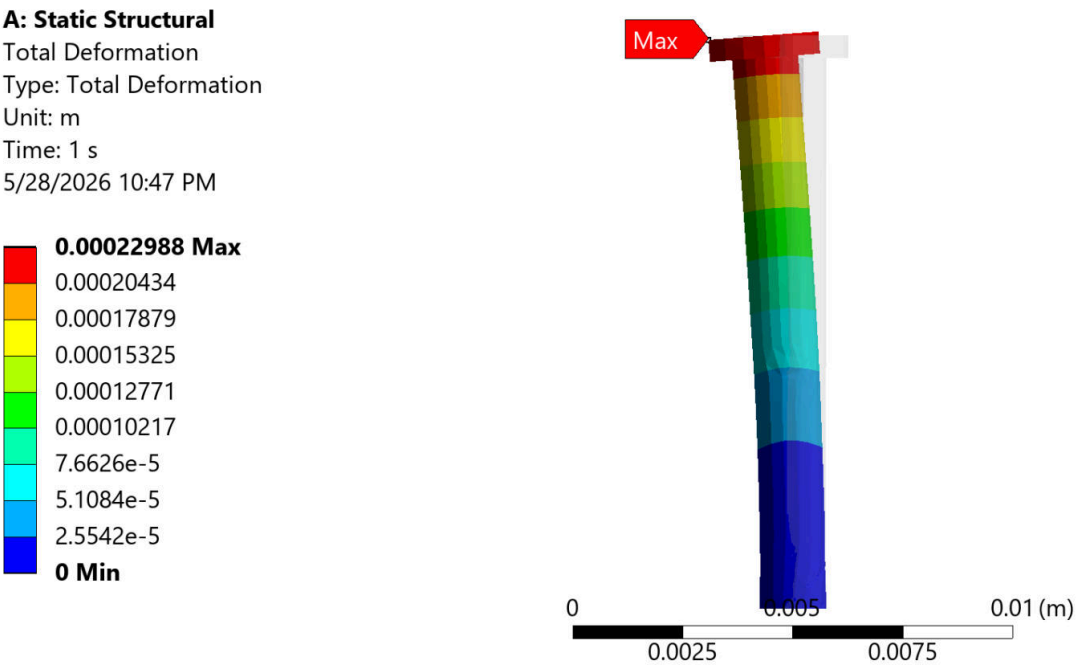
Appendix Figure 1f: 0.0005m Tet10 Total Deformation



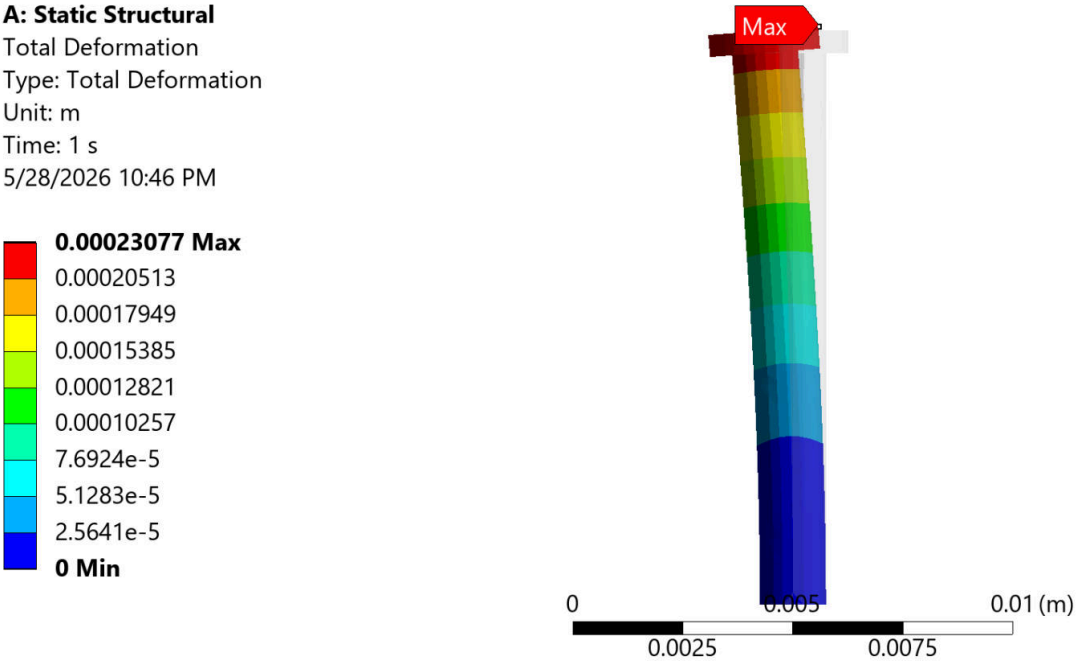
Appendix Figure 1g: 0.0001m Tet10 Total Deformation



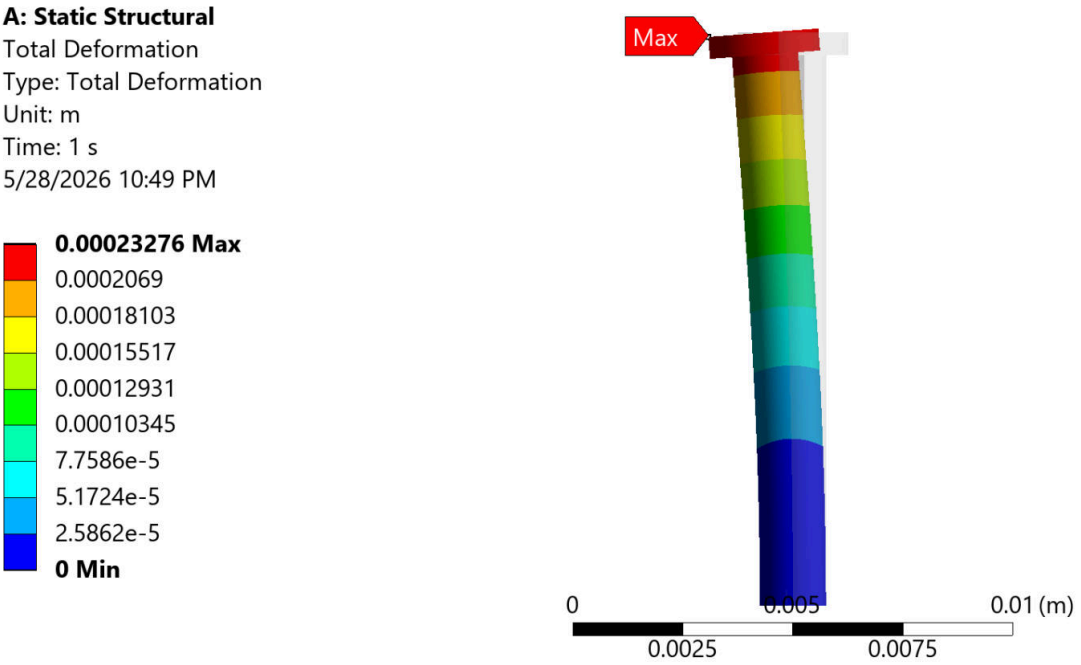
Appendix Figure 1h: 0.00005m Tet10 Total Deformation



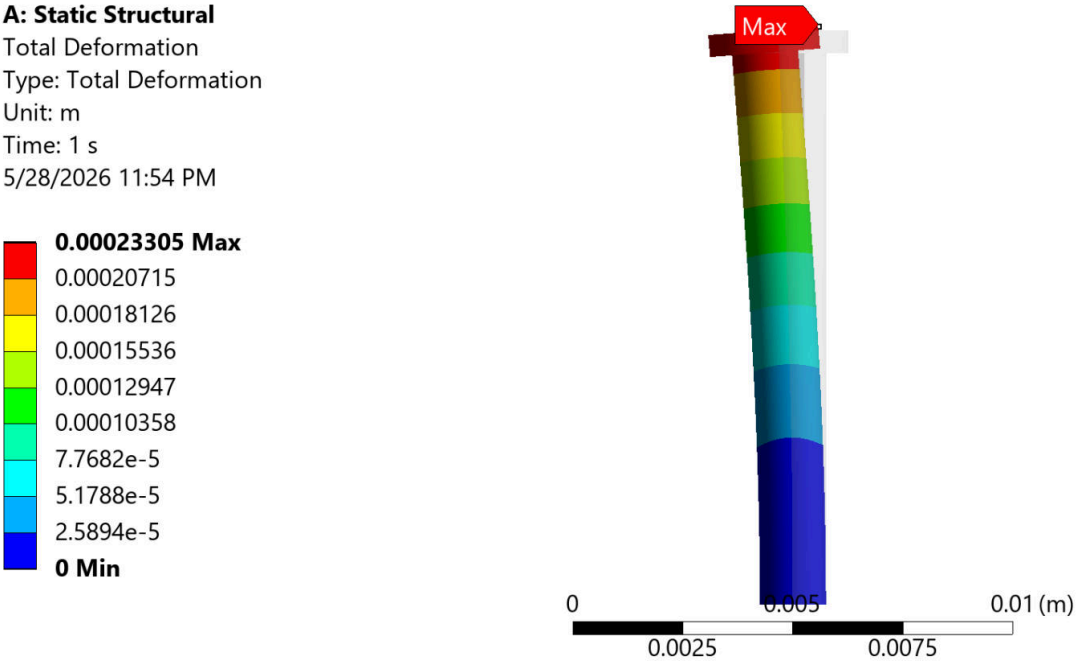
Appendix Figure 1i: 0.001m Hex20 Total Deformation



Appendix Figure 1j: 0.0005m Hex20 Total Deformation

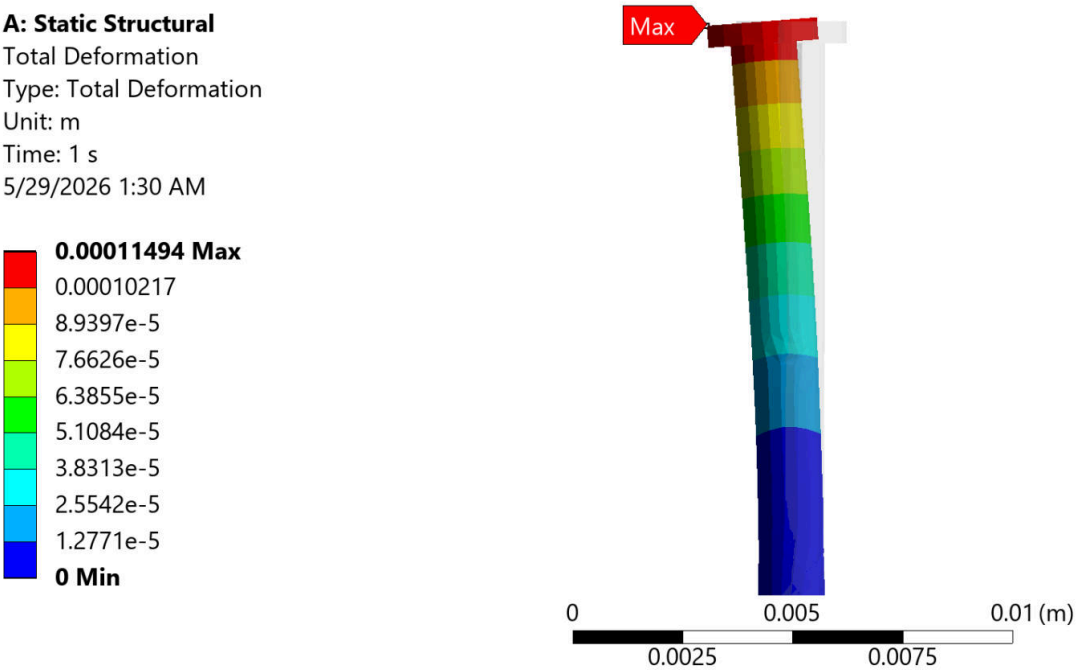


Appendix Figure 1j: 0.0001m Hex20 Total Deformation

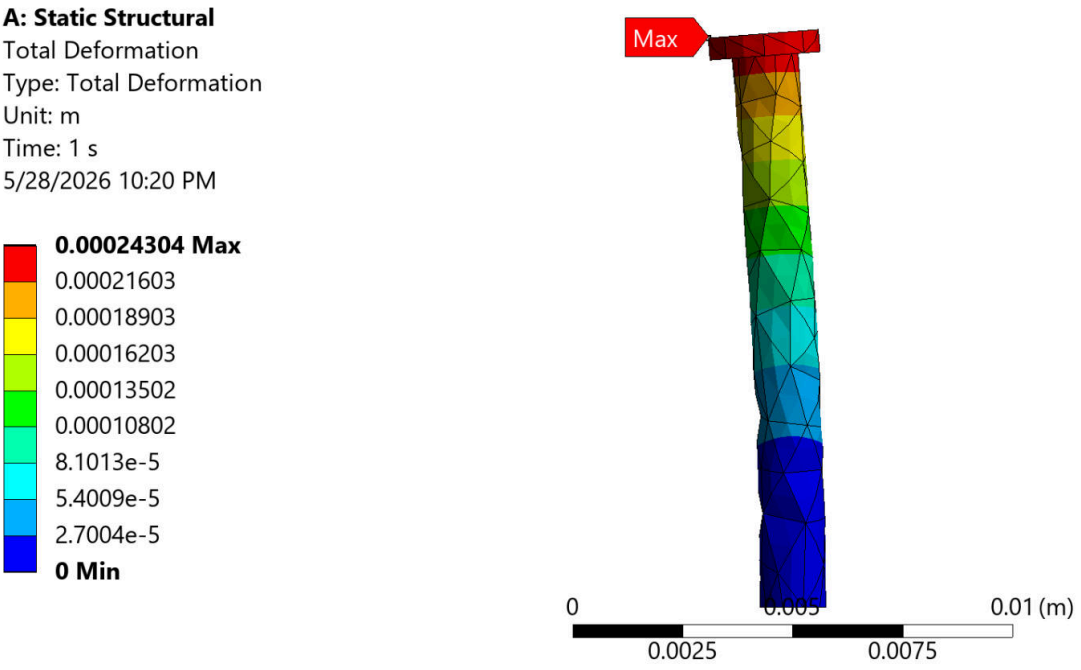


Appendix Figure 2k: 0.000052m Hex20 Total Deformation

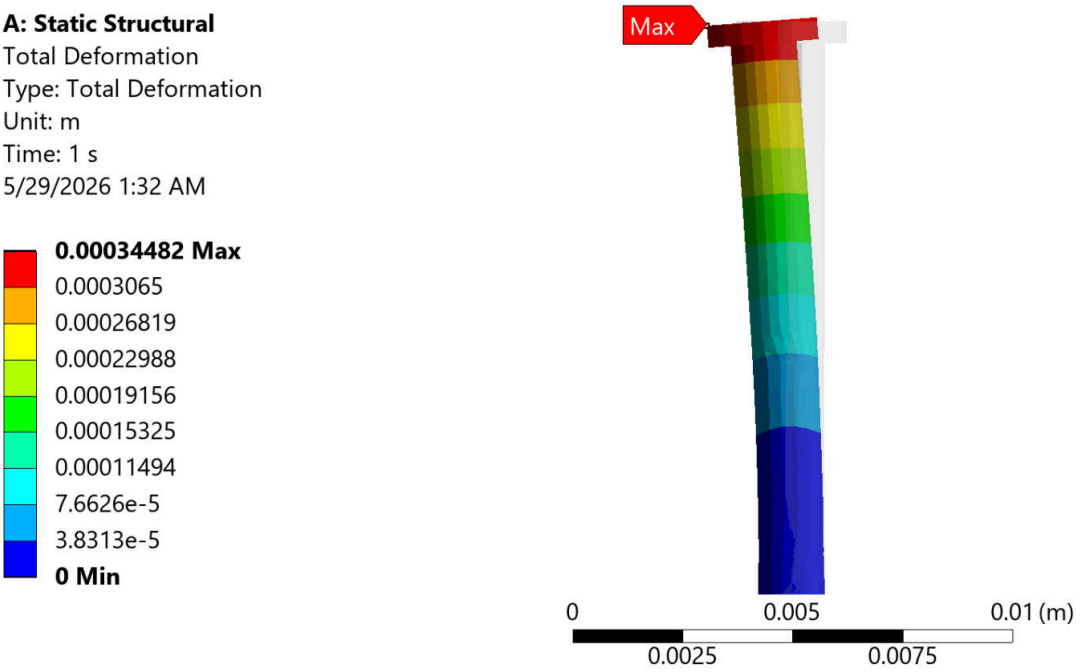
Appendix 3: Spring Constant Total Deformations



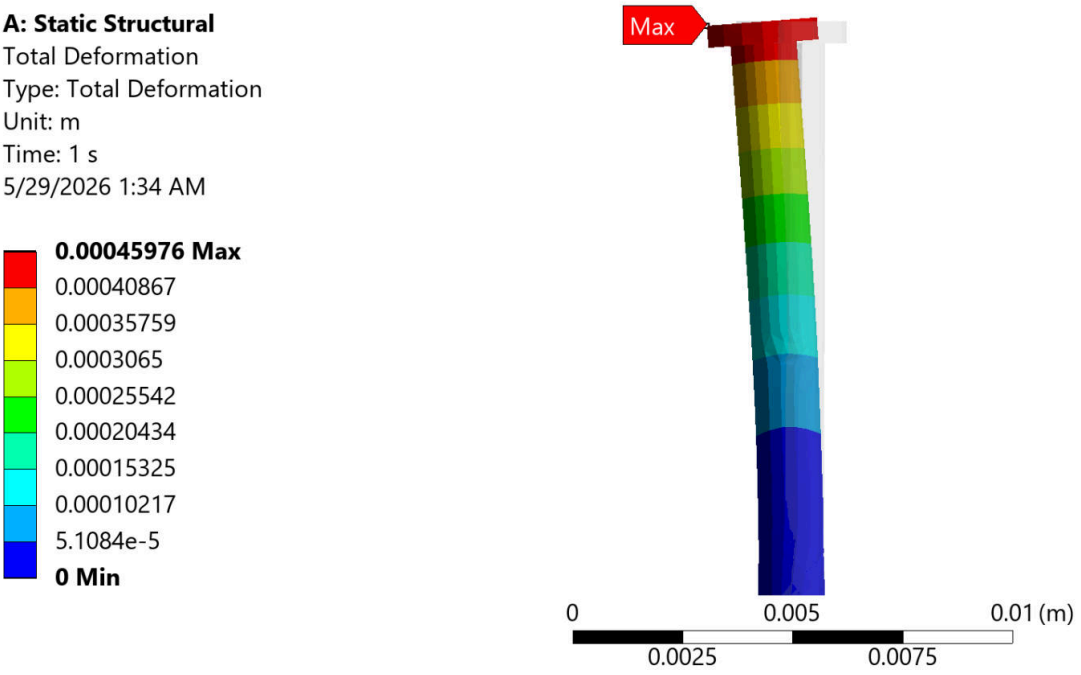
Appendix Figure 3a: 0.001 N



Appendix Figure 3b: 0.002 N



Appendix Figure 3c: 0.003 N

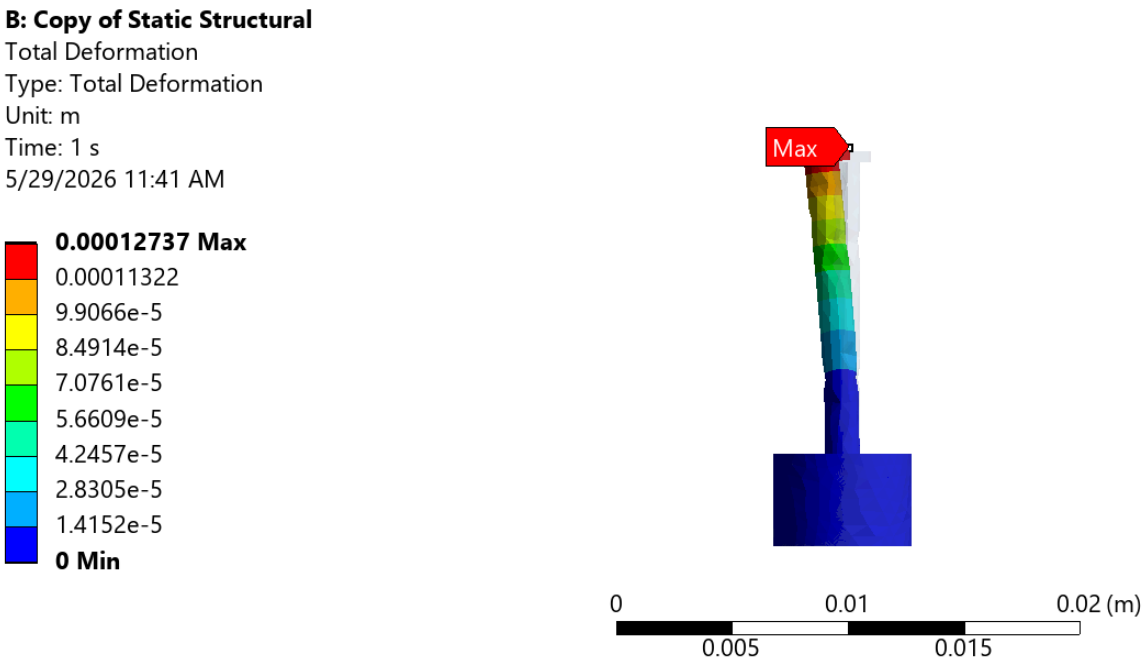


Appendix Figure 3d: 0.004 N

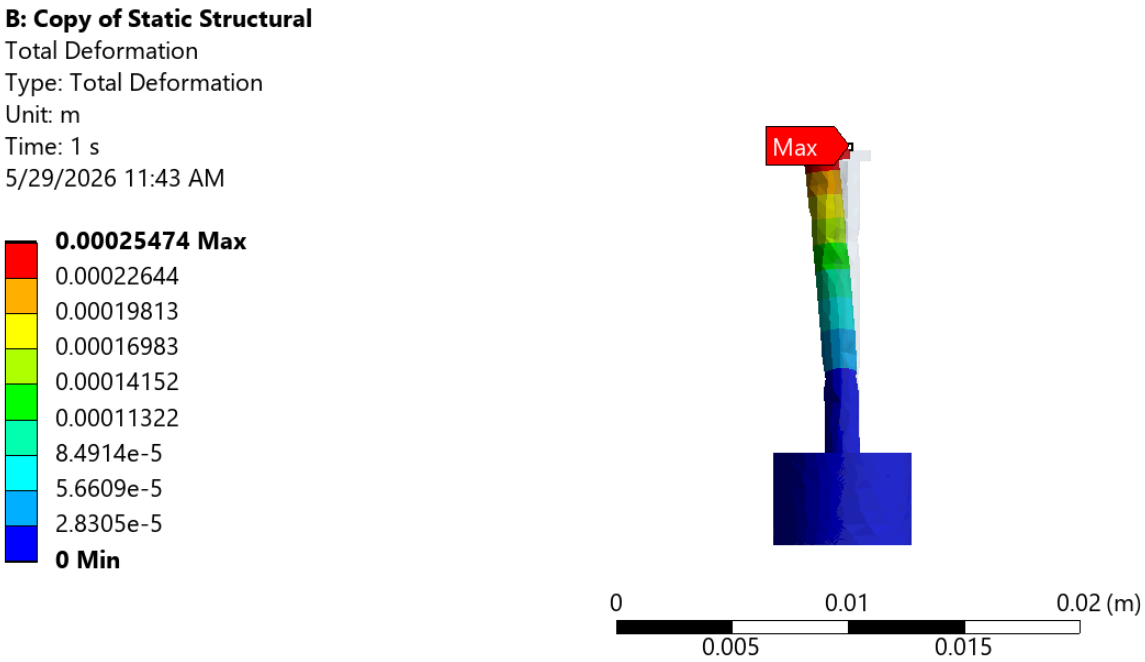
Appendix Table 3a: Results

Force (N)	Maximum Horizontal Deflection (m)
0	0
0.0001	0.00011494
0.0002	0.00024304
0.0003	0.00034482
0.0004	0.00045976

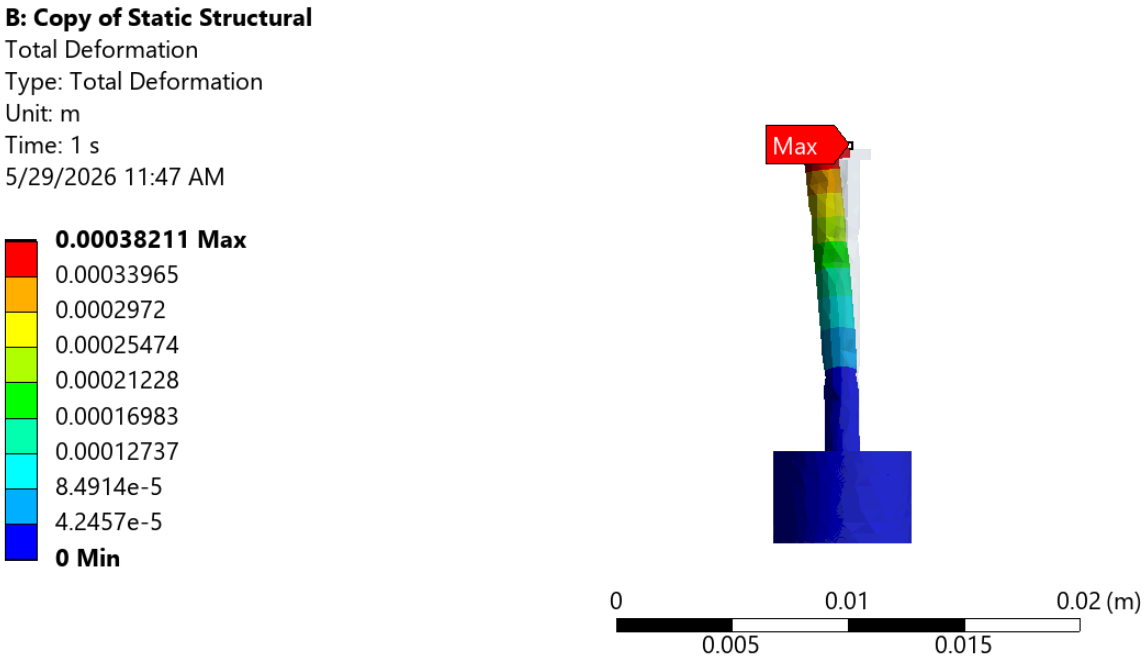
Appendix 4: Spring Constant Total Deformations with Base



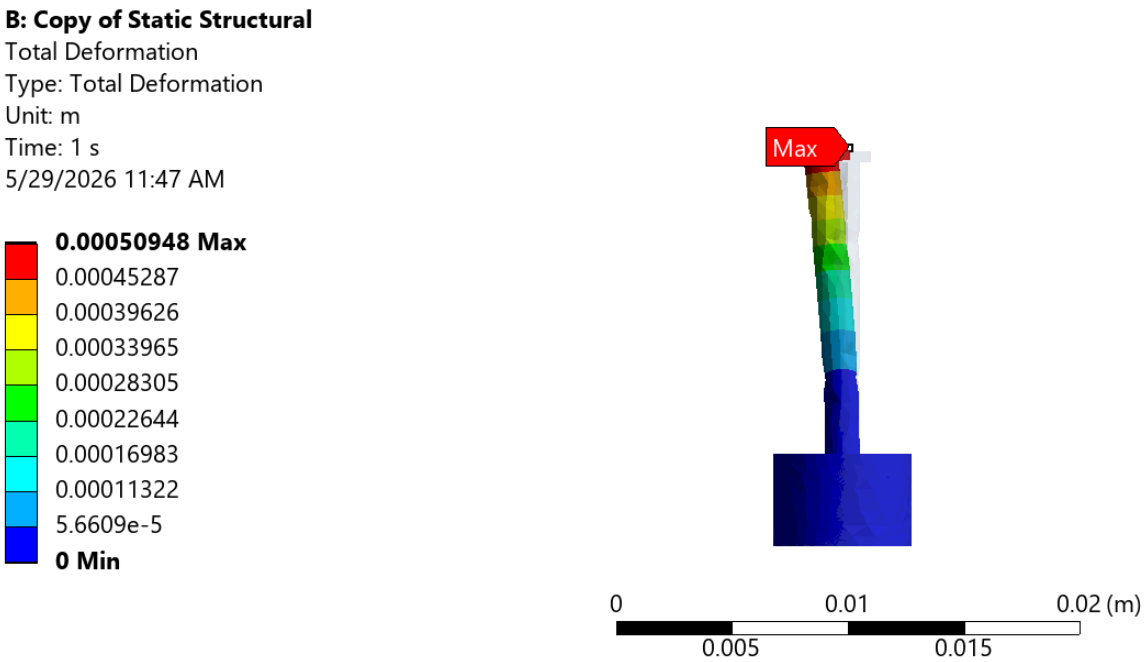
Appendix Figure 4a: 0.001 N



Appendix Figure 4b: 0.002 N



Appendix Figure 4c: 0.003 N

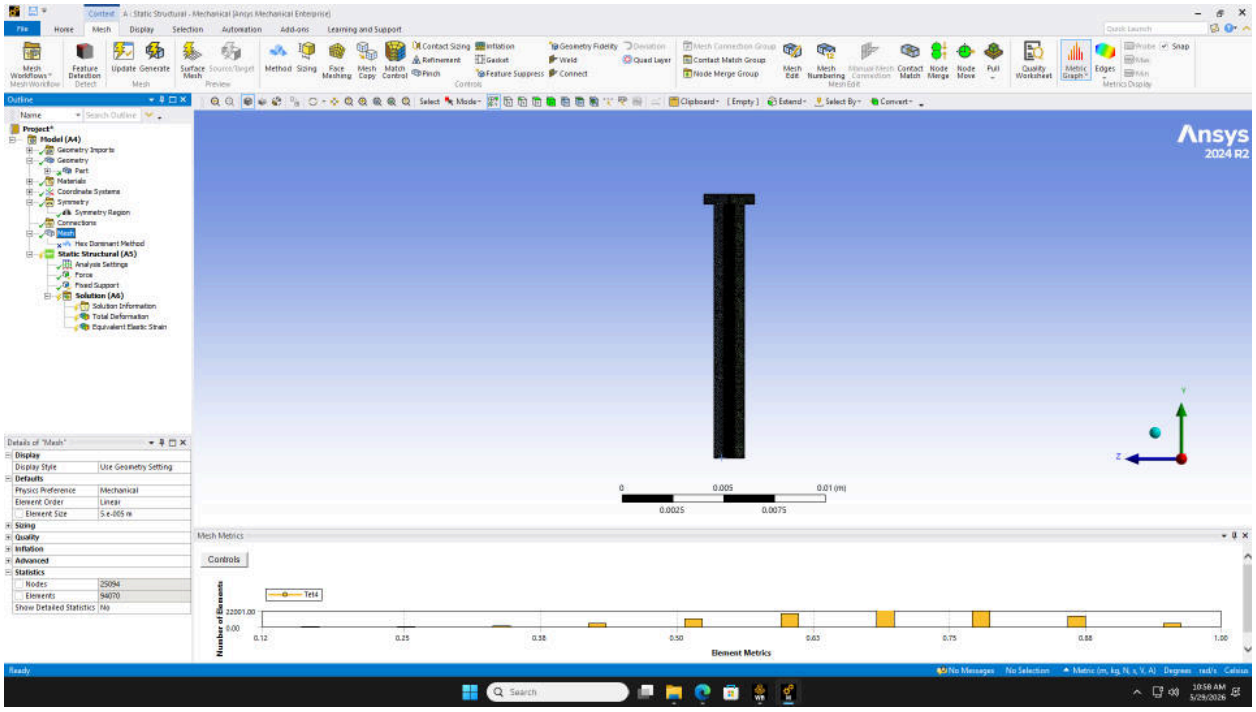


Appendix Figure 4d: 0.004 N

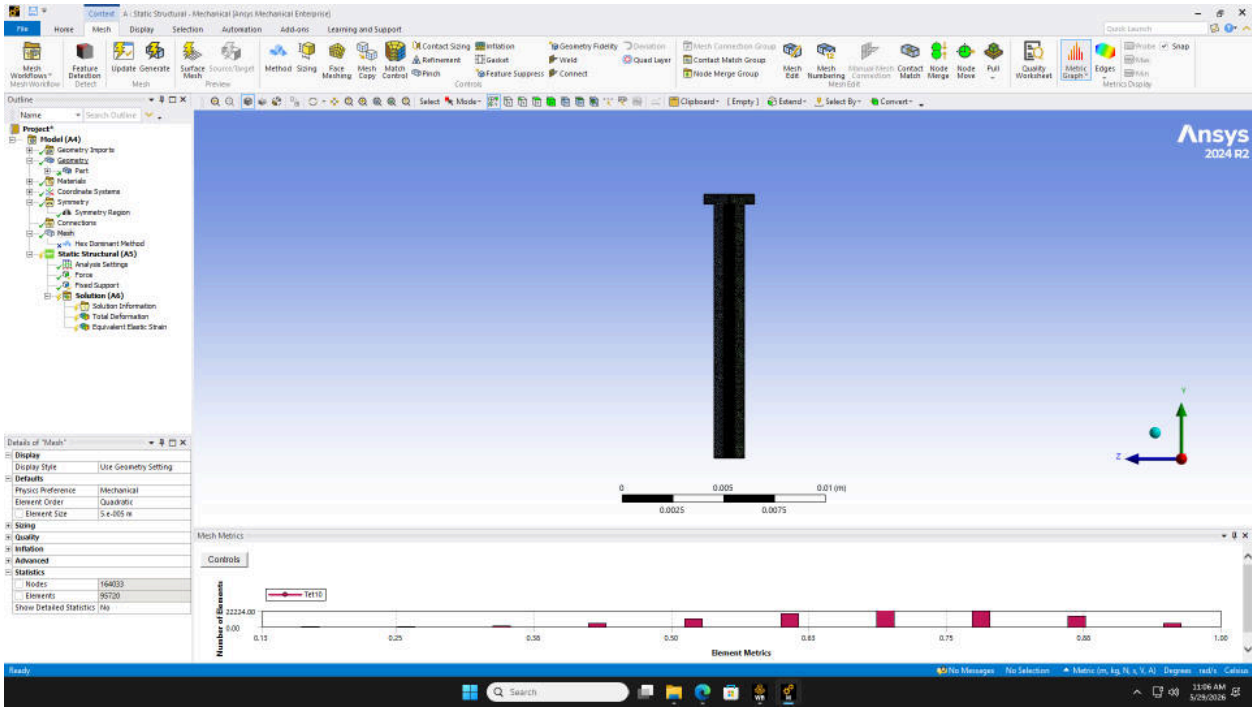
Appendix Table 4a: Results

Force (N)	Maximum Horizontal Deflection (m)
0	0
0.0001	0.00012737
0.0002	0.00025474
0.0003	0.00038211
0.0004	0.00050948

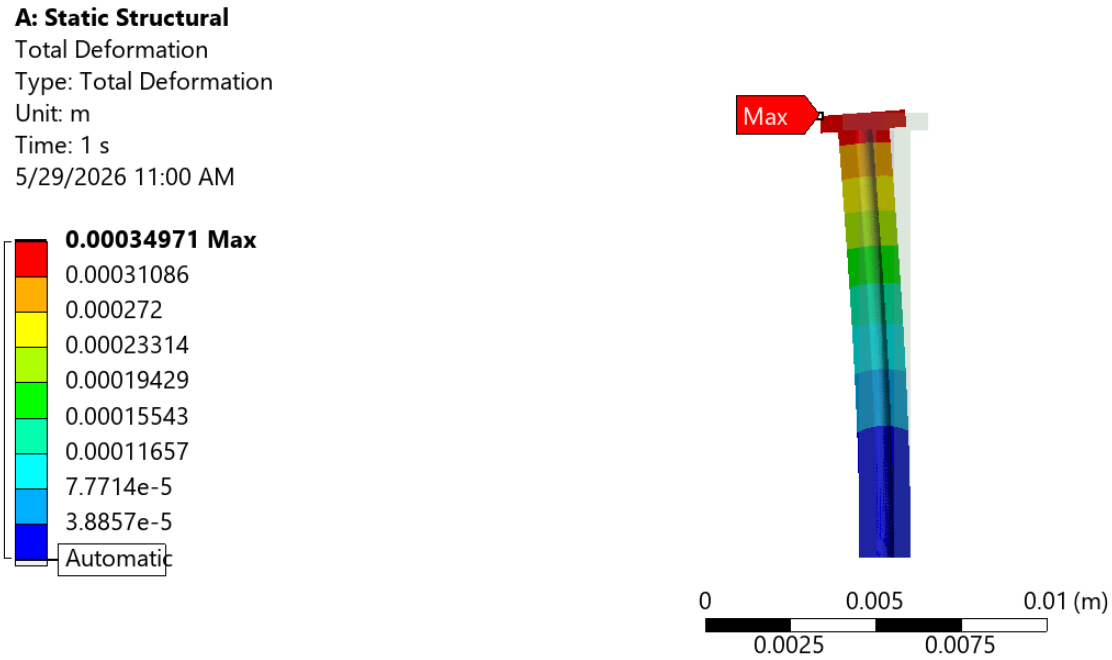
Appendix 5: Unused Mesh and Deformations



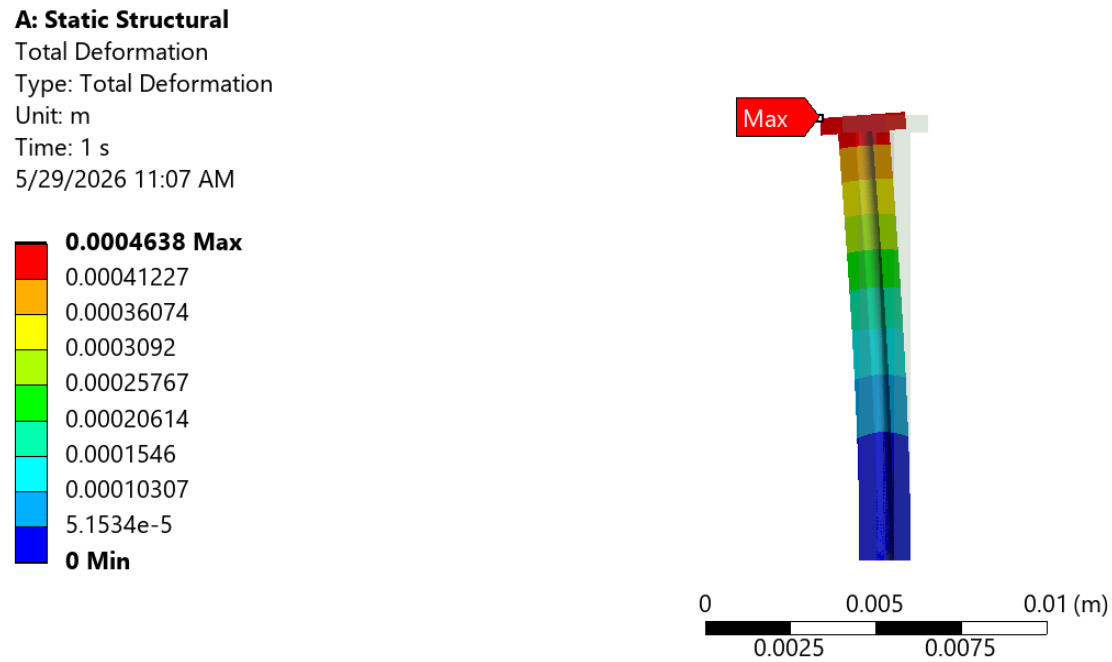
Appendix Figure 4a: 0.00005m Tet4 Mesh



Appendix Figure 4b: 0.00005m Tet10 Mesh



Appendix Figure 4c: 0.00005m Tet4 total deformation



Appendix Figure 4d: 0.00005m Tet10 total deformation

Appendix Table 5a: Results

Element Type	Element Size (m)	Number of Elements	Maximum Horizontal Deflection (m)	Percent Error
Tet4	0.001	280	0.000029673	87.56910009
Tet4	0.0005	632	0.000028751	87.95535324
Tet4	0.0001	20848	0.00030095	26.07688227
Tet4	0.00005	94070	0.00034971	46.50389266
Tet10	0.001	282	0.00051714	116.6452862
Tet10	0.0005	632	0.00044878	88.00725444
Tet10	0.0001	20352	0.00046308	93.99794863
Tet10	0.00005	95720	0.0004638	94.29957798